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School of Engineering

Beyond Clinical

Designing the Next-Generation Mobile Shower Commode Chair for Home-Based Single-Person Care

PAPER WITHIN: *Product Design*

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The authors take full responsibility for opinions, conclusions and findings presented.

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Abstract

This project explores the design of a next-generation mobile shower commode chair for Etac AB, intended for home-based, single-person care within a 10 to 15-year timeframe. Converging trends, such as an ageing population, the sift of healthcare into domestic settings, and rising single-person households will place increasing demands on assistive products to be intuitive, ergonomic, and visually compatible with home environments, a gap that most current products fail to address.

Following the Double Diamond framework, the project combined a STEEP trend analysis, competitor analysis, and user needs mapping to establish 51 product specifications. Six concepts were developed and evaluated through stakeholder workshops at Etac, leading to a final design featuring an organic tube frame, a hyperbolic paraboloid backrest, foldable armrests, and a consistent single-button adjustment mechanism. A four-color palette distinguishes functional zones and highlights interactive elements intuitively.

The final concept was presented as a 1:1 scale physical model and serves as a directional blueprint for Etac's next generation of mobile shower commode chairs.

Sammanfattning

Detta projekt utforskar designen av en nästa generations mobil dusch- och toalettstol för Etac AB, avsedd för hemmabaserad enpersonsvård inom en tidsram på 10 till 15 år. Sammanfallande trender, såsom en åldrande befolkning, förflyttningen av vård till hemmiljöer och ökningen av ensamhushåll, kommer att ställa allt högre krav på hjälpmedelsprodukter att vara intuitiva, ergonomiska och visuellt kompatibla med hemmiljöer. Detta är ett behov som saknas i de flesta av nuvarande produkter på marknaden.

Med utgångspunkt i Double Diamond-ramverket kombinerade projektet en STEEP-trendanalys, konkurrensanalys och kartläggning av användarbehov för att fastställa 51 produktspecifikationer. Sex koncept utvecklades och utvärderades genom intressentworkshops på Etac, vilket ledde fram till en slutlig design med en organisk rörkonstruktion, ett ryggstöd i form av en hyperbolisk paraboloid, uppfällbara armstöd och en konsekvent justeringsmekanism. En pallet med fyra färger särskiljer funktionella delar och framhäver interaktiva element på ett intuitivt sätt.

Det slutgiltiga konceptet presenterades som en fysisk modell i skala 1:1 och fungerar som en vägledande designriktning för Etacs nästa generation av mobila dusch- och toalettstolar

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1 Introduction

1.1 Background

1.1.1 Home-based healthcare

Home-based care is an increasing trend, and demand for it is expected to grow substantially in the coming years (Padyab et al., 2019). This is driven by several factors, one of which is the general ageing of the population, particularly in Sweden (Jalovaara et al., 2019). This will lead to an increase in elderly people in the percentage of the population. Fertility rates across OECD countries have declined from an average of 3.3 children per woman in 1960 to just 1.5 in 2022, contributing to a growing proportion of elderly people in the population (Adema et al., 2024).

1.1.2 Single person households

Trends indicate a rise in single-person households which, combined with growing demand for home-based care, is creating increased pressure on caregiving resources in these settings (Werner et al., 2025). Healthcare products must therefore be intuitive and user-friendly to be operated effectively in contexts where caregivers are understaffed and time-limited with each patient (Sawyer, 1996).

1.1.3 Mobile shower commode chairs

Mobile shower commode chairs are assistive devices used in healthcare to facilitate everyday routines such as showering, toileting, and transportation for individuals with reduced mobility, cognitive challenges, or limited strength. They are used in both institutional settings, such as hospitals and nursing homes, and in home-based care, which requires a degree of versatility in the product (Etac, n.d.). In nearly all cases, the chair is operated by a single caregiver, who uses the primary user's home as their working environment (Tønnessen et al., 2011).

1.1.4 Etac AB

Etac AB is the case company for this project. It is a world-leading developer of innovative assistive equipment designed for people with functional variations. The company is Swedish-based, founded in 1973, and operates offices, manufacturing, and distribution facilities worldwide (AB, 2024).

1.2 Problem description

The shift towards home-based, single-person care places increased demands on newly developed products in the field to be more intuitive, user-friendly, and safe. A mobile shower commode chair should not only be designed with the primary user in mind, but it should also be designed to accommodate caregivers who may not possess detailed

technical knowledge of how more advanced chairs are operated. As future mobile shower chairs are expected to incorporate greater technology, it is important that these technical solutions enhance user-friendliness while also creating a safe environment for the user.

1.3 Objectives and aims

The objective of this project is to design a mobile shower commode chair for Etac intended for use in 10 to 15 years' time. The chair should feature an intuitive design oriented towards single-person care, with a focus on ergonomics and a design language that conveys comfort and long-term use. These aspects are to be grounded in a thoroughly made pre-study examining user needs within single-person care contexts. The design will additionally draw on a trend analysis to identify which shapes and materials may be relevant or prevalent in 10 to 15 years' time. The final mobile shower commode chair is to be presented as a 1:1 scale 3D physical model, accompanied by CAD files and other digital presentations.

The aim of the project is to provide Etac with a blueprint for their next generation of mobile shower commode chairs, which can serve as a foundation for future product development.

1.4 Research questions

Two research questions have been defined to ensure that the study's objectives and aims are addressed in the results.

1. What are the specific requirements for developing next-generation mobile shower commode chair for home-based, single-person healthcare?

2. How can these requirements be translated into the design of a next-generation mobile shower commode chair for home-based, single-person healthcare?

1.5 Delimitations

Several delimitations apply to this project. The project does not address detailed electronic or technical solutions, as this would extend the scope beyond what is feasible. The project does not fully cover manufacturing feasibility of the mobile shower commode chair. The design will therefore not be fully-production ready when presented to Etac.

Ergonomic considerations are based on data available from Etac and anthropometric databases, as formal user testing falls outside the scope of this project.

1.6 Disposition

This project is divided into seven chapters, described as follows:

1. **Introduction** - Describes the background, purpose, objectives, research questions, and delimitations of the project.
2. **Theoretical background** - Presents theories relevant to the field of research, including user-centered design, the Double Diamond framework, and physical ergonomics.
3. **Methods** - Presents the methods used throughout the project to guide the process towards results relevant to product development.
4. **Implementation and results** - Describes how each method was applied across the product development process to arrive at a final concept.
5. **Final concept** - Presents the final concept.
6. **Discussion and conclusion** - Discusses and analyses the results in relation to the research questions, presents conclusions, and suggests directions for further research.
7. **References** - Compiles all sources referenced in the project.
8. **Appendix**

2 Theoretical background

This chapter establishes the theoretical foundation which was used to answer the two research questions.

2.1 Human centered design

Human-centered design is a multidisciplinary paradigm that places the needs, desires, and experiences of people at the center of the development of products, systems, and services, aiming to make them physically, perceptually, cognitively, and emotionally intuitive (Giacomin, 2014).

Designers can at times become preoccupied with their own perspectives and preferences, which may distract from creating solutions that users genuinely find useful and attractive. Effective communication with the user is therefore central to human centered design. Design decisions should be grounded in research rather than assumptions or personal views. At the same time, a design should not blindly follow user requests. The designer's experience and expertise play an important role in interpreting user needs and generating coherent, testable solutions (Boeijen et al., 2020).

2.2 Double diamond

The Double Diamond is a design and innovation framework that is independent of the specific methods used. It helps both designers and non-designers communicate and structure a clear design and innovation process (Design Council, 2023). The framework is divided into four phases arranged in the shape of two diamonds, as illustrated in figure 1.

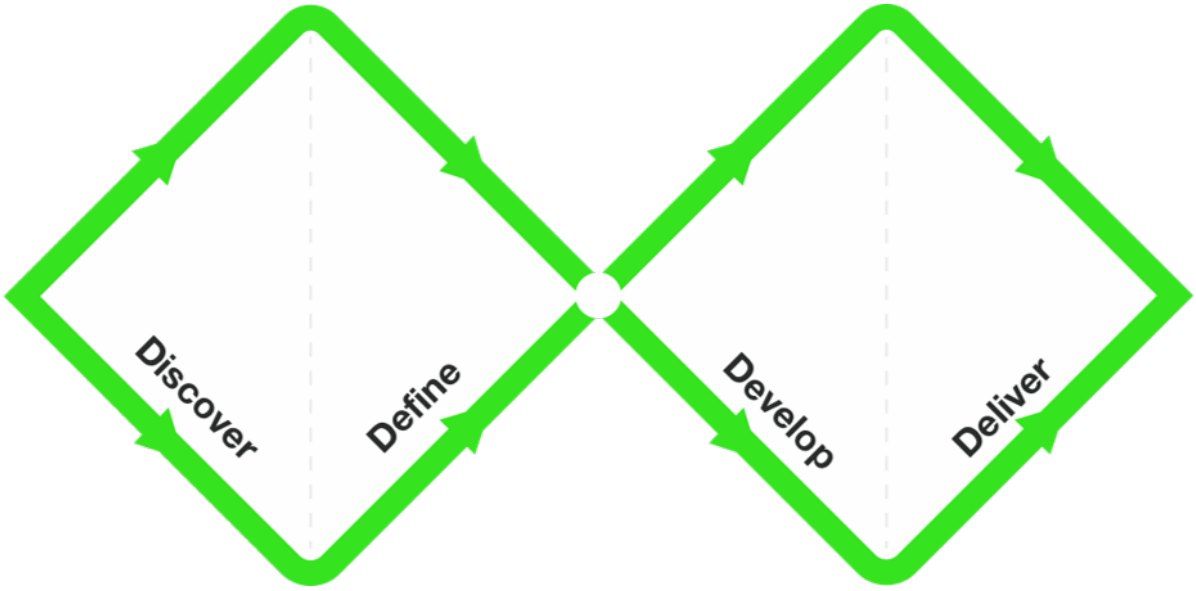


Figure 1: The Double Diamond

The first phase, Discover, is where most of the research takes place. The aim is to gather as much information as possible about the problem and the people affected by it, with a focus on understanding rather than assuming.

The second phase, Define, marks the end of the first diamond. The information gathered during the Discover phase is synthesized and used to articulate new or more clearly defined challenges.

The mindset shifts at the beginning of the second diamond, moving towards convergent thinking. In the Develop phase, ideation is encouraged, generating a wide range of ideas and drawing inspiration from varied sources in response to a well-defined problem.

The final phase, Deliver, involves testing potential solutions, identifying which solutions that work and which do not, and refining the most promising directions while discarding those that are not viable (Design Council, 2023).

2.3 Physical ergonomics

Physical ergonomics is the study of how human body characteristics, such as posture, movement, strength and body dimensions should inform the design of products, workspaces, and tasks to improve comfort, safety, and performance (Bridger, 2018).

2.3.1 Anthropometric ergonomics

Anthropometry is the scientific study of measurements and proportions of the human body (Work, n.d.). Most products require dimensional decisions, such as the sizing of handles and buttons, which are typically based on anthropometric data relevant to the intended user group. In chairs, and especially in medical equipment that must be both functional and comfortable, anthropometric considerations are particularly numerous.

Given the wide variation in human body dimensions across the population, products are generally designed to accommodate a defined percentile range. The most common range is from the 5th to the 95th percentile, covering 90% of the population. In safety-critical products, this range may be extended from the 1st to the 99th percentile (Work, n.d.).

2.3.2 Sitting Ergonomics

Several ergonomic factors must be considered during seated use to maintain function, mobility, and overall health, particularly for users who spend significant periods of time seated. A widely recognized reference point is the "90-90-90" rule, which states that the hips, knees, and ankles should each maintain a 90-degree angle of flexion. This means the thighs should be parallel to the floor and slightly apart, with the feet flat on the ground for support. The pelvis should be symmetrical and level, with weight evenly distributed across both ischial tuberosities. The trunk should remain in the midline,

preserving the natural lumbar lordosis of the spine, as illustrated in Figure 2 (Thirugnanachandran & Bateson, 2012).

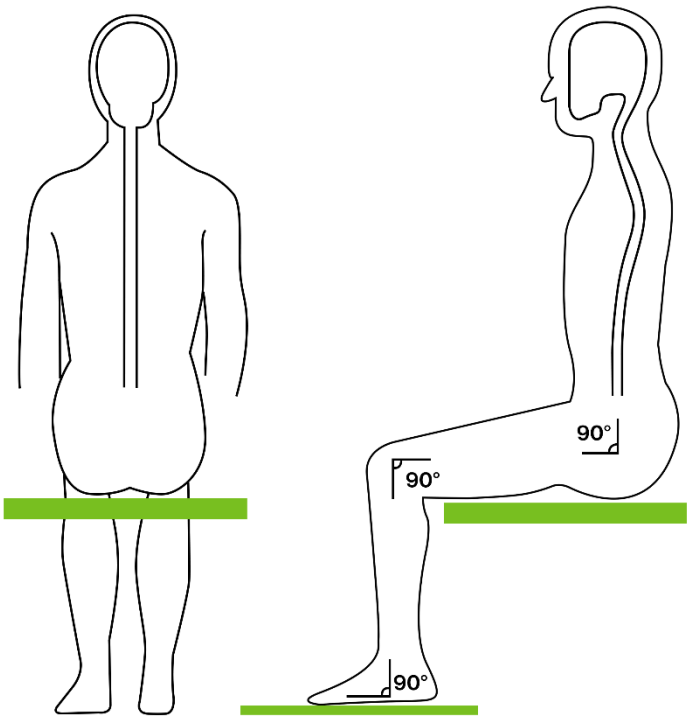


Figure 2: Optimal Seating Position

For elderly users in particular, seating furniture plays an important role in maintaining health, function, and independence, as this group tends to spend a substantial amount of time seated. Seat height should be slightly elevated to facilitate rising from a seated position, and armrests should be stable enough to support the user during this movement (Thirugnanachandran & Bateson, 2012).

Elderly users are also at increased risk of pressure ulcers due to restricted mobility or impaired sensation. Equal distribution of pressure across the seated area is therefore important. A common problem is a tendency to slump into a posterior pelvic tilt, which increases pressure on the sacrum and spine. Conversely, an anterior pelvic tilt increases pressure on the pubis (Figure 3). Both positions elevate the risk of pressure ulcers, and seating should be designed to correct or compensate for these tendencies (Thirugnanachandran & Bateson, 2012).

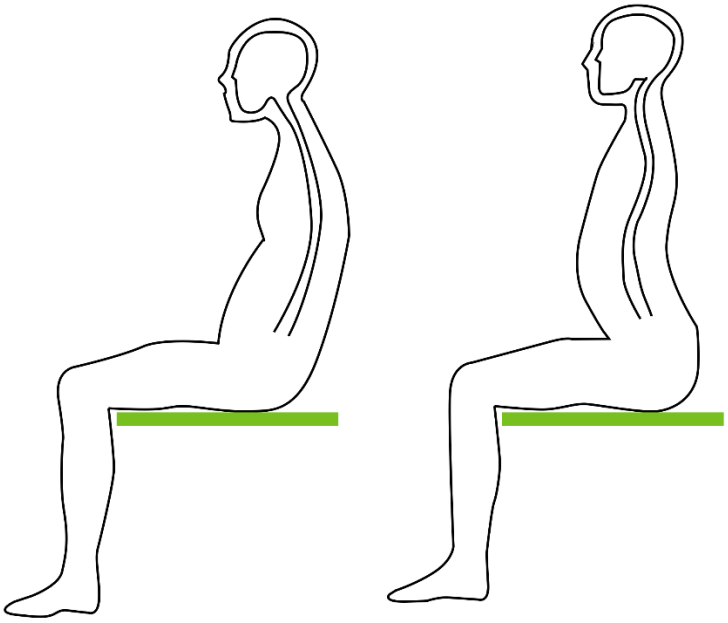


Figure 3: Posterior pelvic tilt on the left and anterior pelvic tilt on the righ

3 Methods

The methods used to collect and analyze data to answer the two research questions are presented according to the double diamond framework.

3.1 Discover

This chapter describes the methods used to collect data relevant to answering the two research questions.

3.1.1 Trend analysis

Trend analysis is a method used to examine how and why things change over time. It can be defined as an analytical approach in which data is gathered to identify patterns and behaviors across different fields (Maggino, 2023). Trends can be categorized by scale into mega-trends, macro-trends, and micro-trends. Mega-trends represent broad, far-reaching forces that influence everyday decisions globally, current examples include demographic change, climate change, and urbanization. Macro-trends emerge as consequences of mega-trends and, while shorter-lived, remain significant, for example, the mega-trend of technological advancement gives rise to macro-trends such as the Internet of Things, smart homes, and artificial intelligence. Micro-trends are outcomes of macro-trends, greater in number but shorter in duration. Different social media platforms, for instance, can be understood as micro-trends under the Internet of Things macro-trend (Figure 4) (globaltrendspotter, 2017).

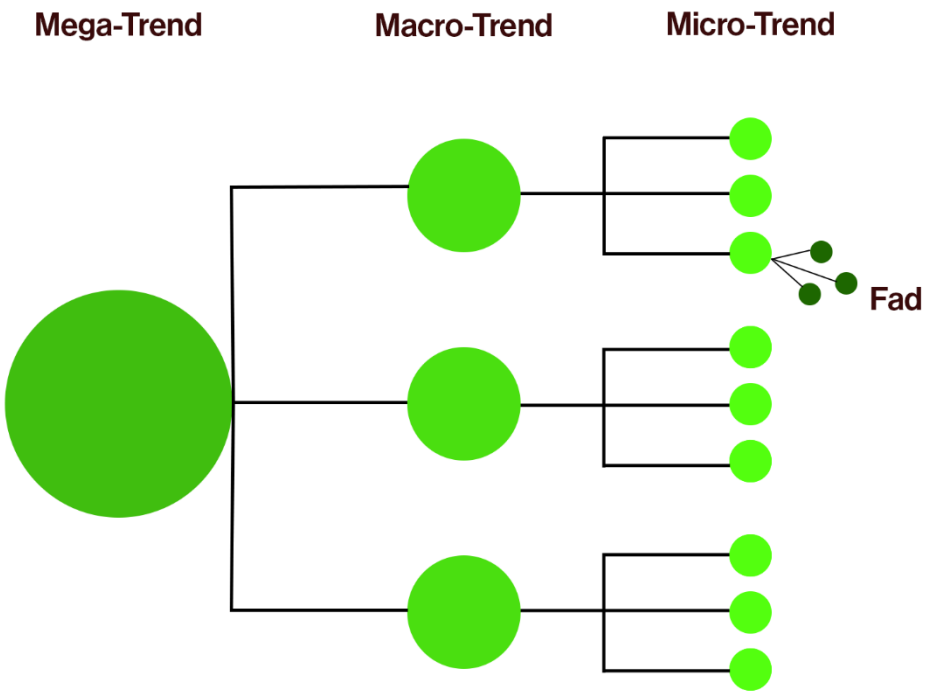


Figure 4: Trend category tree

A further category is the fad, a short-term trend affecting a concentrated area, common in industries such as fashion and food, where interest can peak and fade within weeks. Fads should not be dismissed entirely, however, as they occasionally develop into lasting trends. Social media, initially regarded as a fad, could now almost be considered a mega-trend in its own right (globaltrendspotter, 2017).

3.1.2 STEEP

STEEP is an analytical framework used by organizations to identify, interpret, and monitor trends, forces, and changes occurring outside the organization. The acronym stands for Social, Technological, Economic, Ecological, and Political factors. It functions as a structured checklist for evaluating external factors, such as demographic shifts, emerging technologies, or government regulations, that may create opportunities or threats for business. Given its broad scope, STEEP analysis is an important tool for organizations seeking to anticipate long-term challenges and ensure continued relevance (Fisher et al., 2020).

3.1.3 Competitor analysis

Competitor analysis is a method of examining and evaluating a market segment for a specific product or service based on a defined set of criteria. It is a useful way to identify strengths and weaknesses in the market prior to initiating new design work. A thorough understanding of the product's intended use is essential, as it allows the analyst to assess whether existing competitors have adequately addressed user needs (Milton & Rodgers, 2013). Where gaps exist, there is a strong likelihood that competitors will move to address them, making early identification strategically valuable.

3.1.4 Stakeholder mapping

Stakeholder maps are used to visualize and communicate the key participants involved in a design project. Because relationships among stakeholders evolve throughout the design process, stakeholder mapping is inherently iterative. It is particularly valuable in the early stages to identify individuals and groups who will influence or be affected by the project, typically beginning with a broad brainstorming exercise to determine who may be impacted. Those who stand to benefit from the project are especially important to include, as they often hold decision-making power (Martin & Hanington, 2012).

Once potential stakeholders have been identified, they can be organized according to their roles, creating a hierarchy that clarifies priorities. The resulting map should provide a comprehensive overview of all relevant stakeholders. It may incorporate graphics, text, and photographs there is no single correct format, provided it captures and communicates the key relationships within the design project (Martin & Hanington, 2012).

3.2 Define

This chapter describes the methods used to organize and categorize the collected data into meaningful structures that can be used to address the two research questions.

3.2.1 Personas

Personas is a method used to create fictional characters that represent specific customer profiles for a product or service. Since a product may serve many different types of users, defining who to design for can be challenging. By constructing a character with a fictional name, age, hobbies, family situation, interests, and physical condition, the designer can more easily navigate the exploratory stage of the design process. The persona serves as a representation of the target group, helping the designer understand customer expectations and needs in a cost-effective and straightforward way. Designing for an archetypal user in this manner makes it possible to satisfy a broader group of customers, and the persona can be used as a reference throughout the entire design process (Milton & Rodgers, 2013).

3.2.2 Product specifications

Product specifications are a set list of requirements which spell out in precise, measurable detail on what a product has to do. These are done since user needs generally are less precise and open for too much subjective interpretation. Product specifications do not tell a design team how to address a user's needs but should guide the team in achieving a result which should satisfy the user's needs.

The product specifications should consist of a metric and a value, since they must be measurable. They could be, for example, "average time to assemble", and the according value could be "less than 75 seconds". The value can take on different forms including a specific number, a range, or an inequality (Ulrich et al., 2020).

3.2.3 Mood board

Mood boards are qualitative research tools used in design processes to support creative and innovative thinking. They can serve a variety of purposes, but their core function is to communicate visual ideas by assembling sets of imagery and material samples around a specific theme. Mood boards give designers a structured space to organize visual elements in a way that encourages ideas, inspiration, and creative thinking essential to generating design solutions (Cassidy, 2011).

3.3 Develop

This chapter describes the methods used to generate ideas from the collected data, which will contribute to answering the two research questions.

3.3.1 Brainstorming

Concept generation can take many forms and adhering too rigidly to a structured method risks constraining the creative process. Brainstorming, a tool developed by Osborn in the 1940s, offers a structured yet flexible approach to idea generation that can lead to productive outcomes. Ulrich et al. (2020) describe five guidelines for brainstorming, applicable both individually and in group settings:

- **Suspend judgment**: Unlike routine decision-making, concept generation requires careful consideration, as the consequences of a released product can last for years. Criticism is not permitted during brainstorming unless it leads to new ideas or alternatives.
- **Generate a large quantity of ideas**: A greater number of ideas increases the likelihood that all possible directions have been explored. Ideas also build on one another, so quantity supports quality.
- **Welcome seemingly infeasible ideas**: Ideas that appear impractical can often be refined. Pushing the boundaries of what seems possible may reveal new perspectives or productive combinations with more conventional ideas.
- **Sketch prolifically**: Physical prototyping is time-consuming and verbal descriptions are inefficient. Sketching is a rapid and effective way to communicate ideas and offer new perspectives to others.
- **Build sketch models**: Low-fidelity physical models, made from foam, cardboard, clay, or 3D-printed materials, add depth to ideas. They communicate shape, user interaction, and spatial relationships more effectively than sketches alone, and make it easier for external stakeholders to engage with a concept.

3.3.2 Concept selection

Concept screening is a critical step in product development, as it narrows the solution space and directs resources towards the most promising direction. While organizations use a variety of screening methods tailored to their workflows, this project adopts an external decision-making approach that actively involves stakeholders from the case company (Ulrich et al., 2020).

Each concept is presented alongside a physical mock-up to give stakeholders a tangible reference point. Stakeholders then individually rank all concepts relative to one another. The individual rankings are combined into a total score, and the concept with the lowest total is selected for further development, as this is considered to best reflect the collective assessment of user need fulfilment and overall feasibility.

3.4 Delivery

This chapter describes the method used to present the developed concept in a form that can be used to address the two research questions.

3.4.1 3D Physical Model

Three-dimensional physical models give tangible form to design ideas and are used throughout the design process to evaluate concepts in terms of proportions, aesthetics, and functionality. They can be divided into two broad categories: "looks-like" models, which represent the aesthetic appearance of a design but are generally non-functional; and "works-like" models, which may not resemble the final product but are intended to demonstrate specific functional aspects (Boeijen et al., 2020).

4 Implementation and results

This chapter describes how the methodology was implemented according to the double diamond framework

4.1 Discover

This section describes the results of applying the research methods used to gather information for the project.

4.1.1 Trend analysis

A trend analysis was conducted to gain insight into how healthcare, society, and the broader world look today and how they may develop over the coming years. The analysis was divided into two initial categories. The first examined trends relating to healthcare and society more broadly. The second focused on interior and bathroom design trends, to gather inspiration on how future home environments might look and feel.

The STEEP framework was applied to the first category to ensure that multiple dimensions of societal change were considered. The following trends were identified as relevant to the development of a mobile shower commode chair.

Social trends:

- There is growing public expectation for brands to take responsibility for societal issues, 84% of respondents in VML's survey agreed with this (Chiu & Stafford, 2026).
- Stress is increasingly identified as one of the greatest challenges to human health, with 51% of VML's survey respondents agreeing that it is currently the most significant health concern (Chiu & Stafford, 2026).
- Relationships with AI are becoming a notable social trend (Chiu & Stafford, 2026).
- Users increasingly prefer that assistive equipment does not resemble hospital or clinical devices (Global, 2024).
- Single-person living arrangements are becoming more common (Department, 2024).
- Sweden's population is ageing, with birth rates declining steadily since 1990, as illustrated in Figure 5.

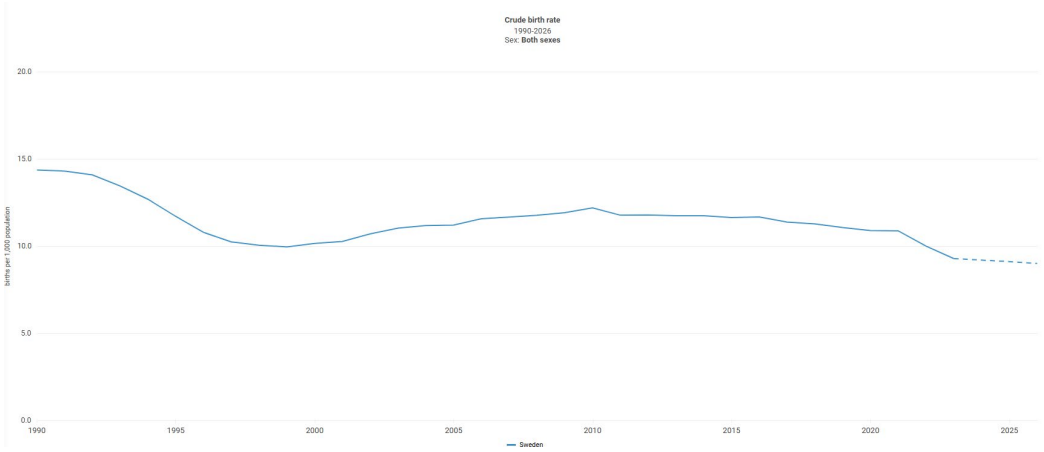


Figure 5: Graph from United Nations data on crude birth rates in Sweden 1990-2026

Technological trends:

- Robots are anticipated to play a greater role in healthcare, 73% of respondents in VML's survey believed that humans and robots will work side by side in the future (Chiu & Stafford, 2026).
- AI-enabled companionship and support tools for people living alone represent an emerging area of development (Wolfe et al., 2025).
- AI is advancing rapidly across multiple sectors.
- Smart living aids, such as toilet reminder sensors, are already entering the market (Huerta, 2025).

Economic trends:

- Ongoing geopolitical instability and conflict may disrupt import and export flows.
- Shifts in global production may affect the cost and availability of components used in assistive devices (Attinasi et al., 2025).
- Material prices may fluctuate because of sanctions or supply chain disruptions.

Ecological trends:

- Consumers expect brands to take responsibility for environmental impact 84% of VML's survey respondents agreed with this (Chiu & Stafford, 2026).
- Color and material trends reflect a broader cultural shift towards natural, muted, and earthy tones, as shown in Figure 7.
- Sustainability is increasingly regarded as a core expectation in product design.

Political trends:

- Healthcare is increasingly shifting away from institutional settings into the home (Bestsennyy et al., 2022).

- Future regulations around sustainability, materials, and responsible production are anticipated.
- Geopolitical conflicts continue to influence trade, sanctions, and the availability of raw materials.

The second category of trends focused on interior and bathroom design. Given that healthcare is moving increasingly into the home, it is relevant to consider how the domestic environments of future users might look. This part of the analysis was conducted without applying the STEEP framework, as interior design represents a niche area where such a broad framework would not be particularly useful. Information was gathered from a range of platforms including social media, design blogs, and forums, and was filtered for relevance to the project.

Interior design trends

The current discourse around interior design for 2026 points towards a significant shift described as "timeless nostalgia", specifically, a revival of 1970s and Art Deco aesthetics. Drawing on reports from Nordic Nest, IKEA, and ELLE, several thematic trajectories emerge:

- **Expressive maximalism**: A move away from minimalist constraints towards personality-driven aesthetics, characterized by playful details and vibrant, joyful color palettes (Sparling, 2025).
- **Artisanal Materiality**: An increased valuation of tactile, handmade elements, including glass decorations, asymmetrical shapes, and intricate textiles such as floral embroidery and statement rugs (Kaminska, 2026).
- **Multisensorial Environments** A holistic approach to space in which visual aesthetics are integrated with auditory and olfactory elements to engage all the senses (Kaminska, 2026; Redaktion, 2026).



Figure 6: AI-generated image of a living room according to the interior design trends using the AI-tool Nanobanana

An AI-generated image (Figure 6) was produced using Nanobanana to synthesize these interior trends into a single visual reference. While it is not straightforward to translate details such as velvet upholstery or woven textiles directly into a mobile shower commode chair, the colors and shapes present in these interiors served as an important source of inspiration during concept development.

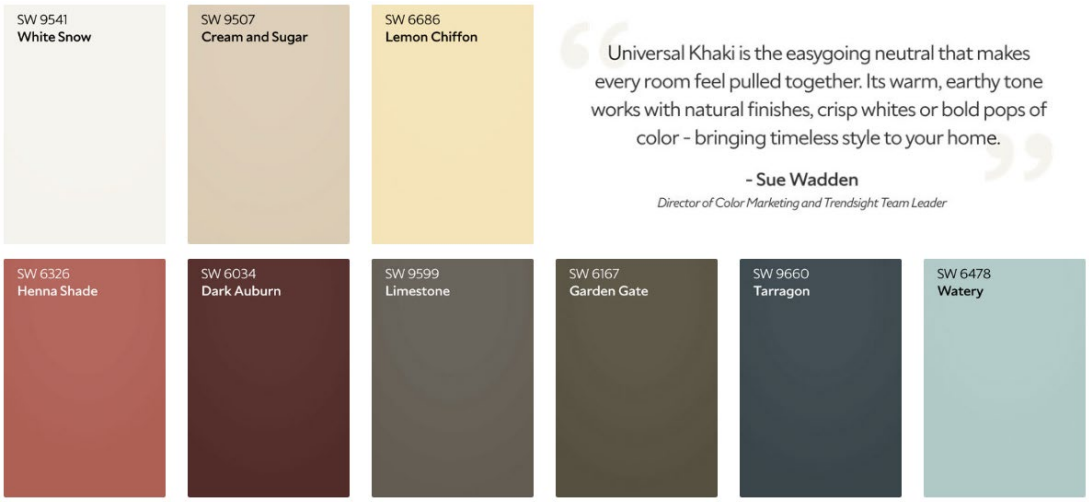


Figure 7: Sherwin Williams color palette for 2026

The Sherwin Williams 2026 color palette (Figure 7) further reflects these trends, featuring warm neutral tones, White Snow (SW 9541), Cream and Sugar (SW 9507), and Lemon Chiffon (SW 6686), alongside deeper, earthier accents including Henna Shade (SW 6326), Dark Auburn (SW 6034), Limestone (SW 9599), Garden Gate (SW 6167),

Tarragon (SW 9660), and Watery (SW 6478). Together, the palette communicates a shift towards timeless, grounded tones that balance warmth with restraint, a natural fit for products intended to integrate into domestic environments.

Bathroom design trends

Contemporary bathroom design for 2026 is oriented around holistic wellbeing, defined by the following key themes:

- **Stylistic evolution**: The rise of "Midimalism" and "Traditional Minimalism" reflects a move away from sterile, clinical environments towards a more "lived-in" aesthetic, balancing minimalist foundations with meaningful additions of color, texture, and personal character (Gustavsberg, 2026; Wells, 2026).
- **Biophilic and restorative design**: There is a strong focus on strengthening the connection between people and nature. Trends such as "Mountain Breeze" integrate modern Scandinavian simplicity with rustic aesthetics, using matte finishes and nature-inspired elements to create restorative, spa-like environments (DECOR, 2024; Gustavsberg, 2026; Wells, 2026).
- **Material and tactility**: Designers are emphasizing authentic organic materials, including light oak, dark walnut, and natural stone such as marble and onyx, alongside tactile surfaces such as fluted or rattan textures, lime plaster, and brushed metal finishes in brass, nickel, and bronze (Gustavsberg, 2026; Wells, 2026).
- **Functional sustainability**: The integration of smart, water- and energy-efficient technology remains a priority, paired with space-saving solutions such as concealed storage and compact modules that optimize limited footprints without compromising aesthetic quality (Gustavsberg, 2026; Wells, 2026).
- **Atmospheric palettes**: Color choices are increasingly influenced by Nordic light conditions, favoring low-saturation earth tones such as greige, sage green, and muted terracotta, with deep blue or black used as structural accents (DECOR, 2024; Gustavsberg, 2026; Wells, 2026).



Figure 8: AI-generated image of a bathroom according to the bathroom design trends using the AI-tool Nanobanana

An AI-generated image (Figure 8), produced using Nanobanana, was created to summarize these bathroom trends visually. As with the interior trends image, this does not represent a direct blueprint for the final chair design but rather serves as a condensed visual reference of the collected trends used to guide concept ideation.

4.1.2 User needs and product specifications

Since the mobile shower commode chair is an assistive device, the user must remain central to all design decisions. User needs were gathered through a secondary data collection process conducted by the case company. These were subsequently discussed with Etac employees, structured, and organized in a way that made them applicable to this project. The user statements collected by the company (Figure 9) reflect the perspective of the caregiver, who is identified as the primary user of the product, given that it is the caregiver who physically operates the chair. These statements were reformulated into user needs following the method described by (Ulrich et al., 2020).

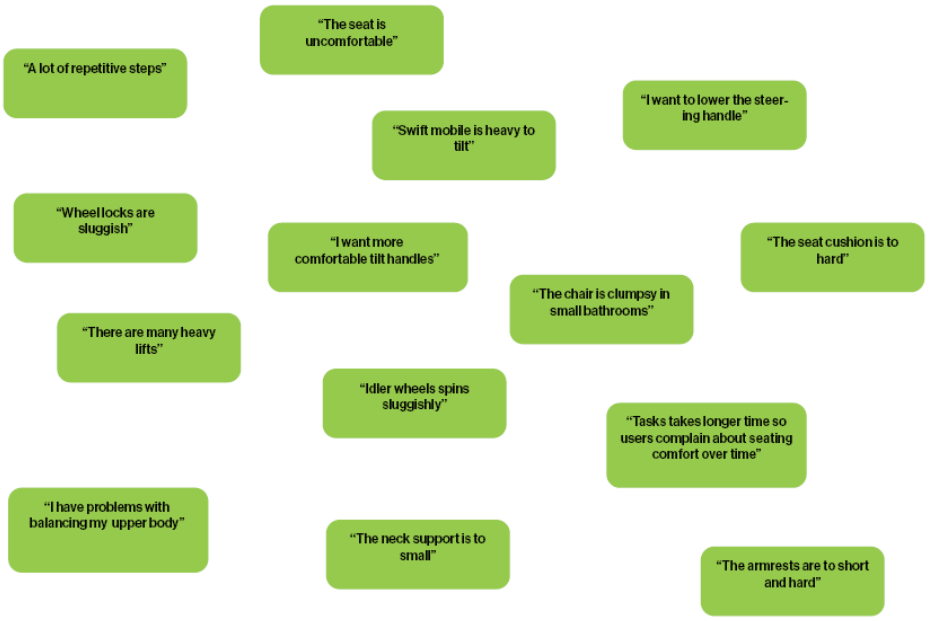


Figure 9: Some of the user statements given by Etac

Etac also provided a set of desired specifications for the chair, which were assessed for relevance and integrated into the project accordingly. The identified trends were likewise translated into user needs and then into specifications, contributing to a final list of 51 product specifications (see Appendix 1) that informed the concept development process.

Many of the metric-based specifications relate to dimensional requirements that can be verified during 3D modelling. Others require physical user testing or formal certification testing, which falls outside the scope of this project and therefore cannot be guaranteed. The following test procedures are proposed as examples of how selected specifications could be verified in a subsequent development phase:

R04 – Maximum safe working load Place a calibrated dead load of exactly 150 kg on the seat and leave it for a minimum of 30 minutes. Inspect all structural joints, the frame, and the seat for permanent deformation, cracking, or fastener failure. Repeat the test at 176 kg (representing a 10% safety margin) and apply the same inspection criteria to confirm the safety margin and maximum safe working load.

M04 – Rotation within defined space Mark a circle of 990 mm diameter on a flat floor using tape. With a test occupant seated at the 50th percentile weight, a second person performs a full 360° rotation while holding the chair's handles. The specification is passed when the full rotation is completed without any part of the chair crossing the boundary of the marked circle.

H03 – Disinfection resistance Using a standard hospital-grade disinfectant wipe, perform 500 full wipe-down cycles across all exposed surfaces, concentrating on the

armrests, backrest, and seat. After every 100 cycles, photograph and inspect surfaces for discoloration, surface cracking, material swelling, or delamination. The specification is passed when no functional or cosmetic degradation is observed after the full 500 cycles.

U02 – Intuitiveness for first-time users Recruit five participants with no prior experience with the product. Provide them with only the standard user documentation, no verbal instruction. Time each participant from the moment they begin reading until they can independently and correctly perform the core operations: adjusting the backrest, neck rest, and footrest, and locking all wheels. The specification is passed if all five participants complete all tasks within 120 seconds.

C03 – Noise during operation in a room with a confirmed background noise level of ≤25 dB, position a calibrated sound level meter at seated ear height, 0.5 m to the side of the chair. Perform a standard maneuver sequence, 2 m forward travel, a 360° rotation, and wheel lock engagement and release, with a 75 kg test load. The specification is passed if no single reading exceeds 30 dB at any point during the sequence.

4.1.2.1 Product specification ranking

Given the number of specifications identified, a prioritization method was applied to ensure appropriate weighting during concept development. The MoSCoW method was used to rank specifications based on Etac's initial priority designations (Figure) and subsequent discussions about how well each specification aligned with the project's objectives. The full ranked specification list is provided in the Appendix.

etac
Creating Possibilities

DESCRIPTION	REQUIREMENT	MOSCOW
Standard	Fulfill ISO 17966	Must have
Service life	High quality - durable Lifetime 10 years	Should have
Material	Biocompatible	Must have
	Sustainable	Should have
Technical	Central locking wheels	Should have
	Integrated technology	Should have
	Ergonomical both for user and caregiver	Must have
Usability	Single-use care	Must have
	Fool proof design	Should have
Cleaning	Surface easy to clean/wipe off	Must have
Washing	Waterproof (no water absorption allowed)	Must have
	Cleaning in 85 °C (for 3 min without discolouration and/or change in shape)	Should have
Disinfection	Desinfection tolerant	Must have

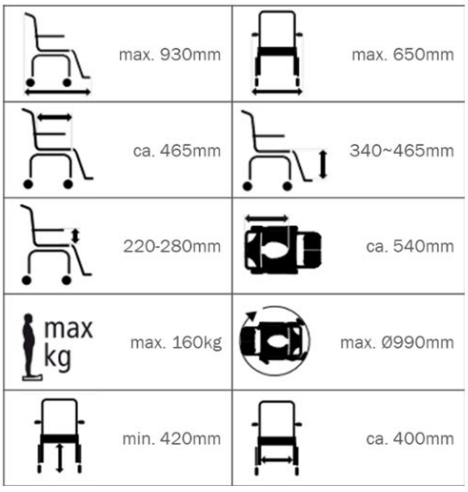


Figure 10: Product specifications given by Etac

4.1.3 Competitor analysis

A competitor analysis was conducted to evaluate existing variations of the mobile shower commode chair. Twenty chairs from different brands were identified through online research and evaluated against the product specifications and user needs in terms of both function and aesthetics. No geographical restrictions were applied, as a broader market view was considered more beneficial than limiting the analysis to European brands alone. The chairs evaluated are shown in Figure 11.



Figure 11: Mobile shower commode chairs evaluated in the competitor analysis

The most significant structural observation from the competitor analysis is the near-universal use of stainless-steel tube construction, well-illustrated by the Drive Healthcare Wheeled Commode TRS 130 (Figure 12). This tube-based frame typically takes on a cubic form, which strongly defines the aesthetic of most chairs currently on the market. The remaining components, such as seat pads, armrests, and backrests, are generally manufactured from molded polypropylene or polyurethane foam, both of which offer adequate resistance to the chemicals and water exposure inherent in this type of product.



Figure 12: Driven healthcare Wheeled Commode TRS 130

The overall aesthetic of existing chairs is clearly shaped by their institutional context. The colors and geometric forms are well suited to hospital or clinical environments, but it is evident in many of these designs that aesthetic considerations were not a primary concern. For a product increasingly being used in people's homes, this represents a notable gap in the current market.

4.2 Define

This section describes the results of applying the methods used to define and structure the research gathered for this project.

4.2.1 Persona

Two personas were developed to represent the two main users of the mobile shower commode chair, the primary user (the person who sits in the chair) and the caregiver (the person who operates it). Both personas were informed by conversations with Etac employees regarding the typical profiles of their product users.

The first persona, Ove Nilsson (Figure 13), represents the primary user. Ove is not a perfect or universal representation of this user group, as primary users can vary considerably, from individuals managing chronic illness at a relatively young age to elderly people who find daily tasks such as showering or toileting increasingly difficult due to age-related physical decline.

It is important to note that the primary user is not the most directly considered user in the design of the mobile shower commode chair. The significant variation between individuals makes it difficult to design a single chair that precisely addresses the needs of all primary users. This contrasts with products such as custom wheelchairs, which are configured to order because users depend on them continuously. The commode

chair, by comparison, is used less frequently and in more standardized ways, which shifts the design focus towards the caregiver.



Figure 13: Persona of the main user

The second persona, Therese Fransson (Figure 14), represents the caregiver. While this profile is also a simplification, caregivers tend to be a more consistent user group in terms of how they interact with the chair. Therese operates the chair independently in time-pressured environments, where efficiency and safety must be balanced simultaneously.

Etac identifies the caregiver as the primary design target for the mobile shower commode chair. As the person responsible for operating the chair, the caregiver must be able to understand and manage its mechanisms, including adjustments and wheel locking, to provide the best possible experience for the primary user. As a result, most chair design decisions are oriented towards the caregiver's needs rather than those of the primary user.



Figure 14: Persona of the caregiver

4.2.2 Stakeholder map

A stakeholder map was developed to define the range of people involved in the project, with particular attention to the end users (Figure 15). As illustrated, a significant number of parties are involved in the lifecycle of an assistive product of this kind.

In Sweden, products distributed through the healthcare system are procured by the individual healthcare regions, which set technical and functional requirements through formal procurement processes. The design of a Swedish mobile shower commode chair is therefore largely shaped by the requirements established in these procurements, and development cycles are often tied to upcoming procurement rounds.

Since primary users in Sweden receive their chair on loan from their healthcare region, they do not select the product themselves. The choice is typically made by an occupational therapist or a relative. Nevertheless, the end users, including the primary user, the caregiver, and the user's family, remain the ultimate target group and are central to the objectives of this project.

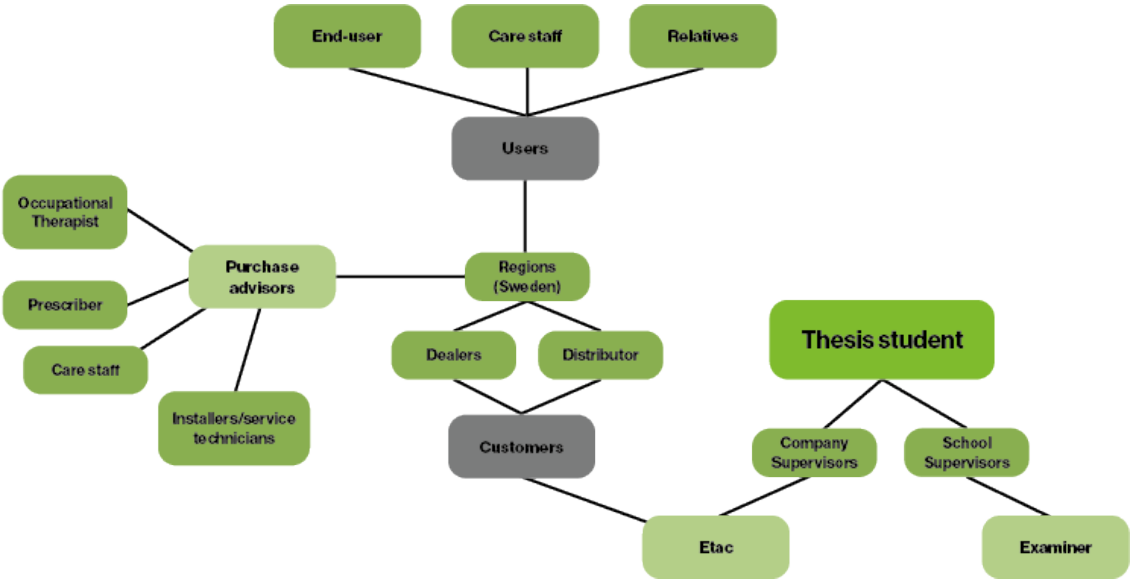


Figure 15: Stakeholder map

4.2.3 Mood- and color board

Two mood boards were developed to guide the visual direction of the mobile shower commode chair design, one addressing the overall feelings the design should convey, and one focusing on the color palette to be incorporated.



Figure 16: Mood board describing overall feeling and shape inspiration for the mobile shower commode chair

The mood board shown in Figure 16 communicates the intended emotional qualities of the design. The decision to use organic shapes was directly informed by the trend analysis, which identified biophilic, restorative aesthetics as a dominant direction in future bathroom design (Gustavsberg, 2026; Wells, 2026). These organic shapes are intended to help the chair integrate naturally into domestic bathroom environments, while also aligning with Etac's existing product portfolio, which favors rounder, softer forms. Bent cylindrical details help to unify the structural elements of the chair, while curved surface areas serve both ergonomic and aesthetic purposes, reinforcing the biophilic quality of the overall design.



Figure 17: Color board

Four colors were selected for the chair, shown in Figure 17. The first three were drawn from the Sherwin Williams 2026 color palette, selected for their alignment with the broader trends identified in the analysis, calm, restorative tones that evoke Nordic light and create a relaxing environment. The fourth color, RAL 2004 (a vivid orange), was assigned specifically to the adjustable components of the chair. Using a single, consistent accent color across all interactive elements creates a visual system that communicates adjustability intuitively, if the caregiver recognizes one adjustment point, they can identify the rest by the same color cue.

4.3 Develop

This section describes the results of applying the methods used to generate and refine design ideas in response to the project's findings.

4.3.1 Brainstorming

The initial concept generation phase began with hand sketching using paper and pen, which allowed for rapid, low-commitment exploration of two-dimensional forms (Figure 18). Given the complexity of the mobile shower commode chair, which comprises many individual parts and functions, the early sketching sessions focused on establishing the overall character and silhouette of the chair rather than resolving specific mechanical details. The goal at this stage was to depart significantly from existing designs,

particularly those identified in the competitor analysis, and to explore what a fundamentally different structural language might look like.

Photoshop was subsequently used as a digital sketching tool, enabling cleaner linework and easier color application, which proved useful when presenting concepts to project stakeholders (Figure 19).

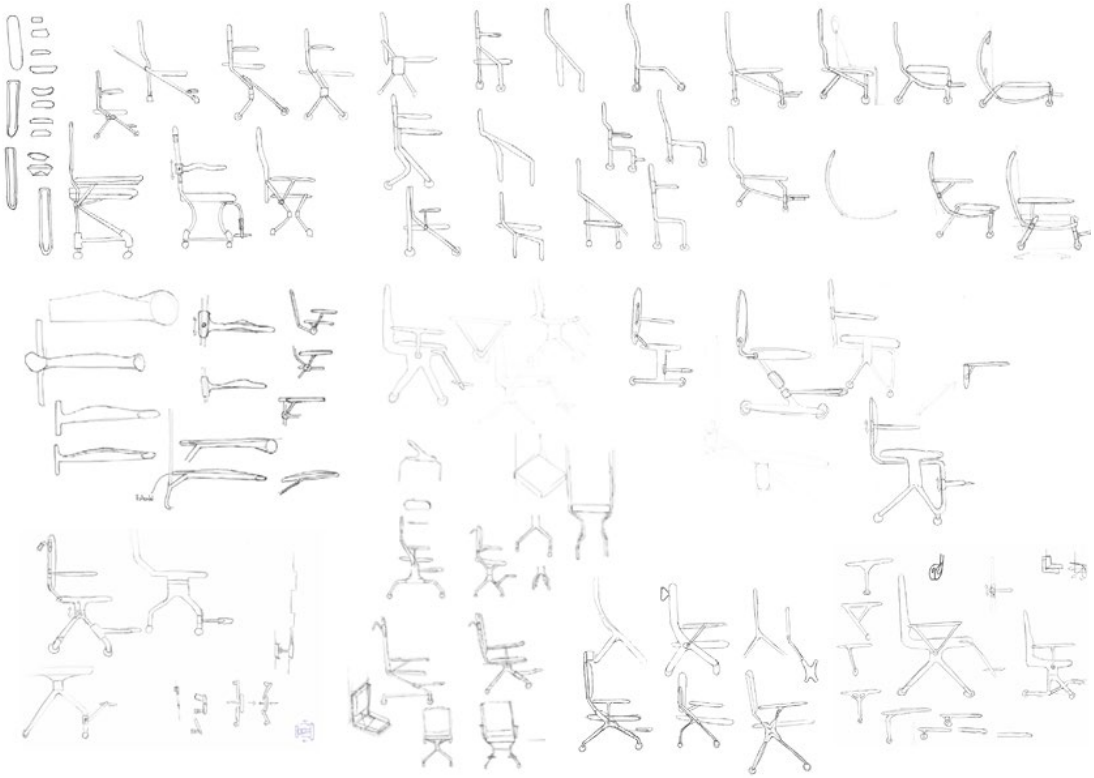


Figure 18: Hand sketches during brainstorming

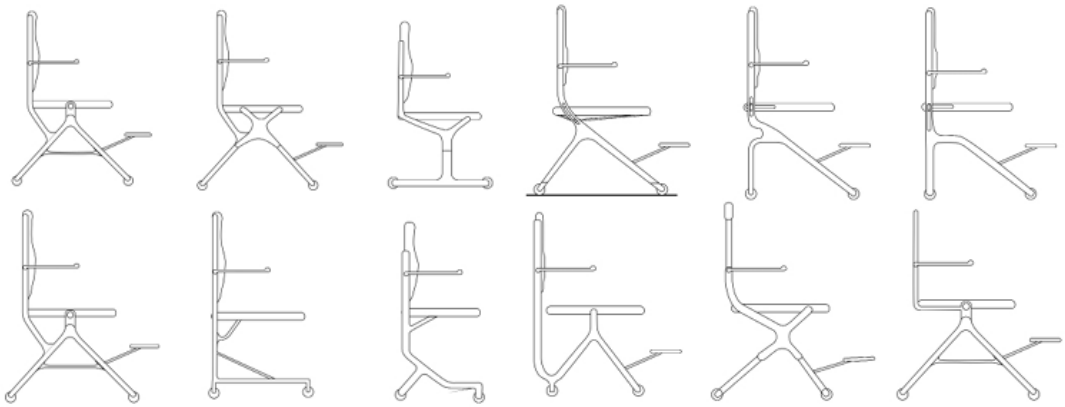


Figure 19: Hand sketches during brainstorming

A key design constraint that shaped all concepts from the outset is the requirement for caregiver accessibility. The primary user cannot be enclosed within the chair structure, as this would prevent the caregiver from repositioning or assisting the user as needed. A clear gap must therefore be maintained between the backrest and the seat, and the

sides of the chair between the armrests and the seat must remain open. While these constraints are difficult to fully communicate in two-dimensional sketches, they were considered throughout the brainstorming process and became more clearly resolved as the work transitioned into three-dimensional modelling.

Six distinct concepts were developed in 3D, each described below.

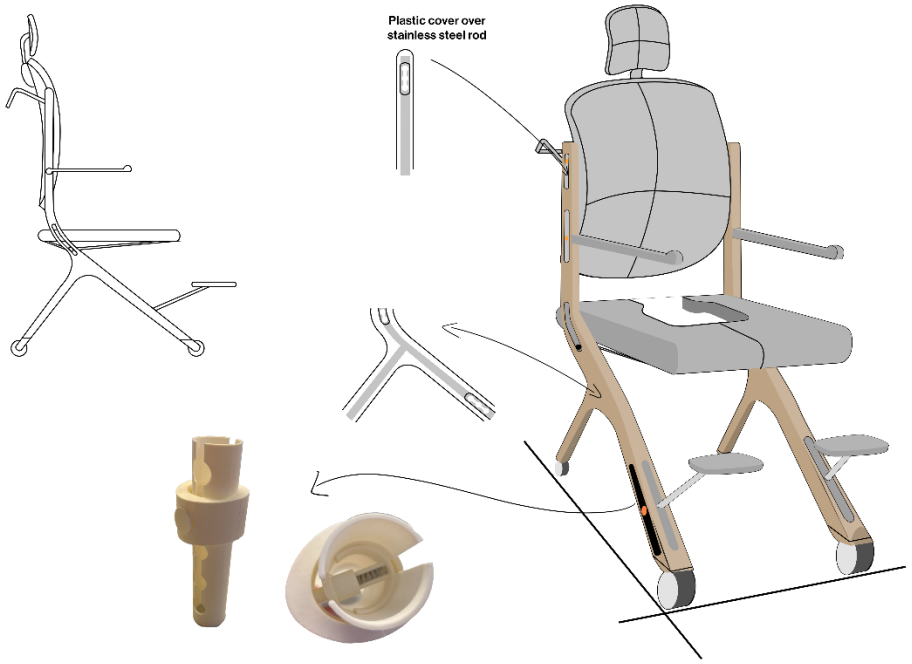


Figure 20: First Concept



Figure 21: AI contextual render of the first concept

Concept 1

(Figure 20) The first concept explores a chair composed of two mirrored two-dimensional side profiles with a turning, organic form, unified and clean character. Each side was envisioned as an inner stainless-steel tube enclosed within a plastic cover, removing the exposed-tube aesthetic common in existing designs. The cover also conceals the four adjustment mechanisms located on each side, which govern the footrests, seat, armrests, and caregiver handlebars. A consistent adjustment mechanism across all four points was a deliberate decision, if the caregiver understands how one adjustment works, they can operate all of them without additional learning.

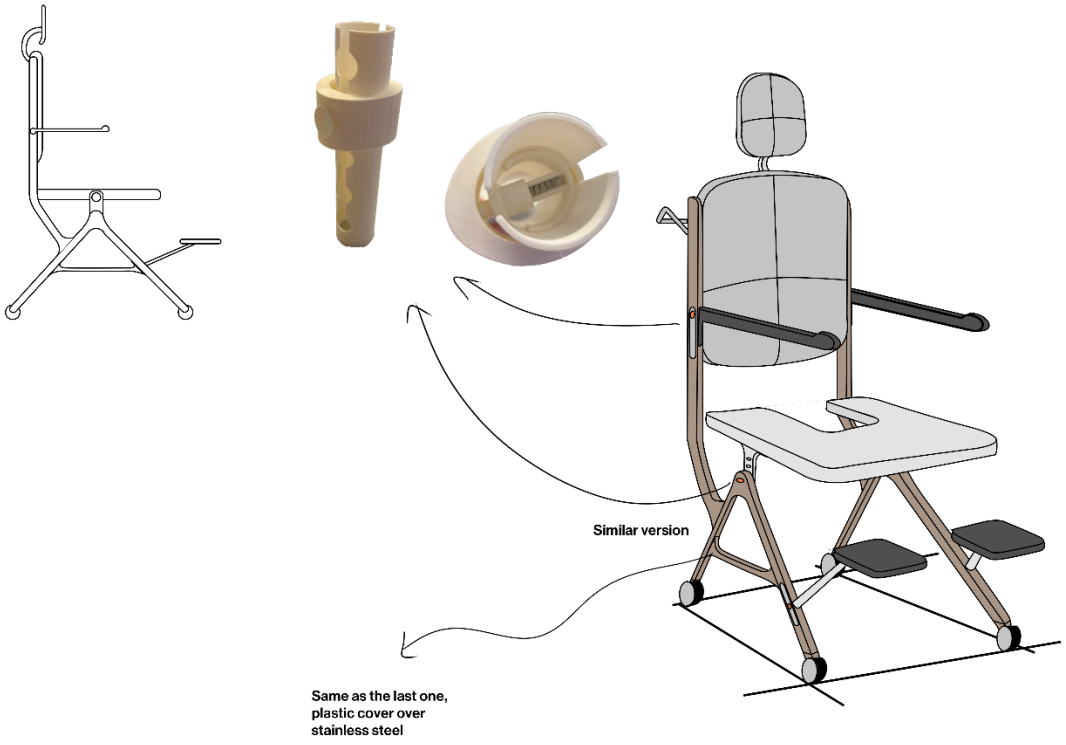


Figure 22: Second concept



Figure 23: AI contextual render of the second concept

Concept 2

(Figure 22) The second concept is closely related to the first but introduces an "A"-shaped leg configuration, where the backrest uprights branch from the leg structure like limbs from a tree trunk. The seat height adjustment mechanism in this version benefits from more centered load distribution, improving stability. The overall form is somewhat more complex than Concept 1, however, which was noted as a potential drawback.

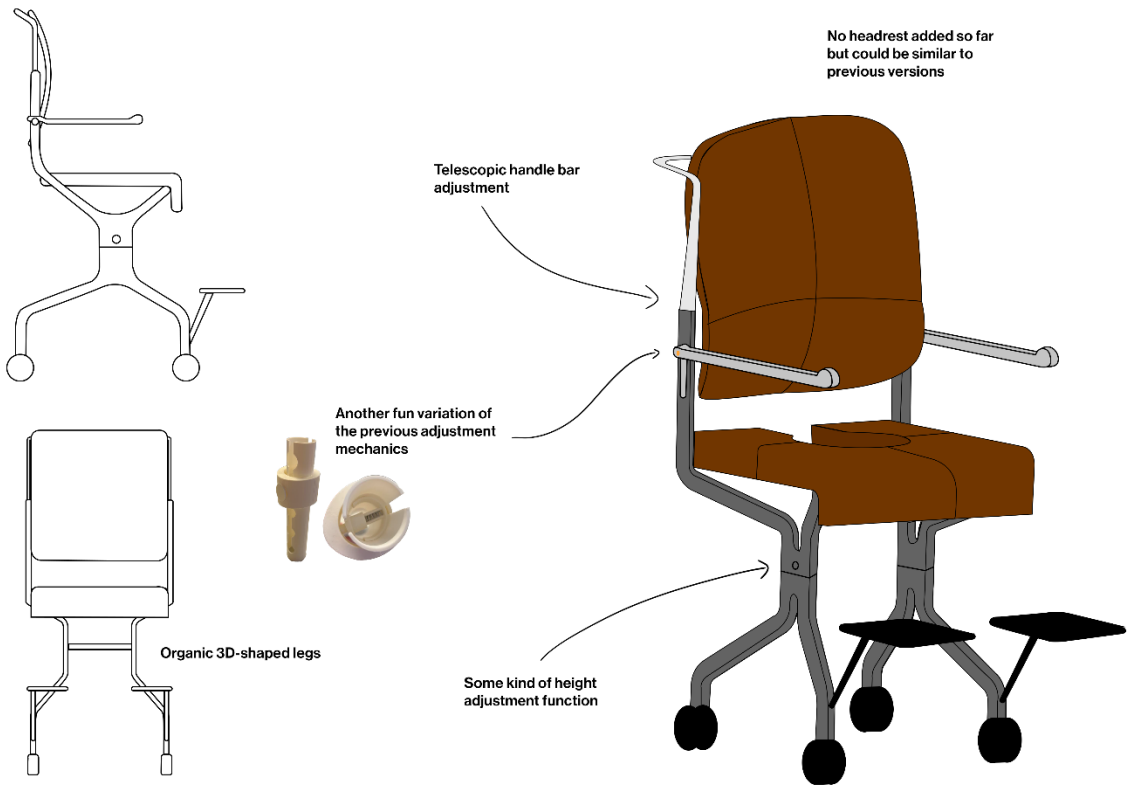


Figure 24: Third concept



Figure 25: AI contextual render of the third concept

Concept 3 (Figure 25) The third concept introduces three-dimensional bending to the side frame elements, giving the chair a more distinctly organic character. The rear of the side frame curves around the seat, while the front merges beneath it, creating an integrated relationship between structure and seating surface. The caregiver handlebars feature a telescopic adjustment mechanism that introduces a material or

color transition, adding visual interest. Armrests and footrests use the same adjustment approach as in the previous concepts.

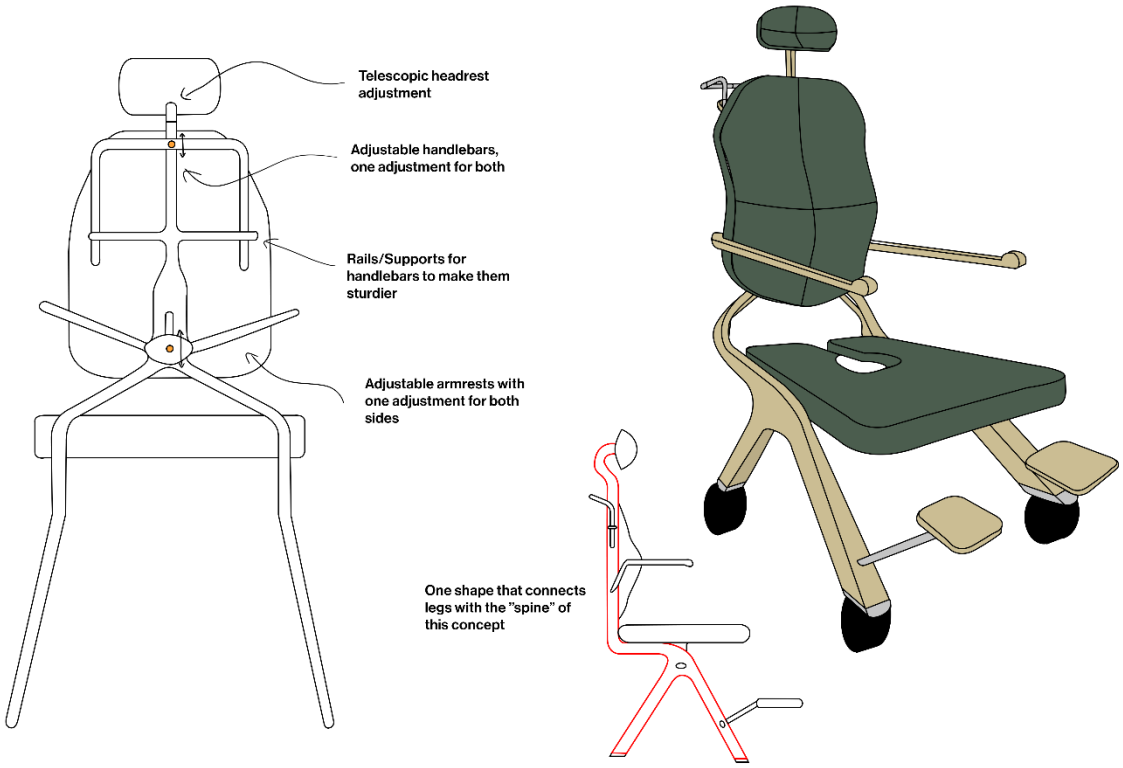


Figure 26: Fourth concept



Figure 27: AI contextual render of the fourth concept

Concept 4 (Figure 26) The fourth concept departs from the conventional two-sided frame by merging the side elements into a central backbone running behind the chair. This backbone serves as the mounting point for multiple functions, the armrests and

caregiver handlebars each connect to a single shared adjustment point, reducing the total number of required interactions and saving time for the caregiver. The design language aims for a frame that reads as a single, continuous organic form, moving away from the angular, segmented appearance of current market products.

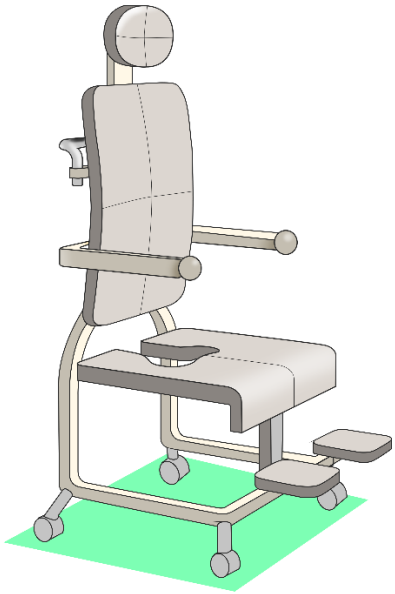


Figure 28: Fifth concept



Figure 29: AI contextual render of the fifth concept

Concept 5 (Figure 28) The fifth concept is a more restrained variation of Concept 4, with a less organic form and a fixed front section. The footrests are also mounted centrally at the front and adjust vertically from a single point, further reducing the number of individual mechanisms. The more rectangular lower structure lends itself particularly well to the integration of a central wheel-locking system, a feature absent from most current designs, which require each wheel to be locked individually, at a cost to efficiency.

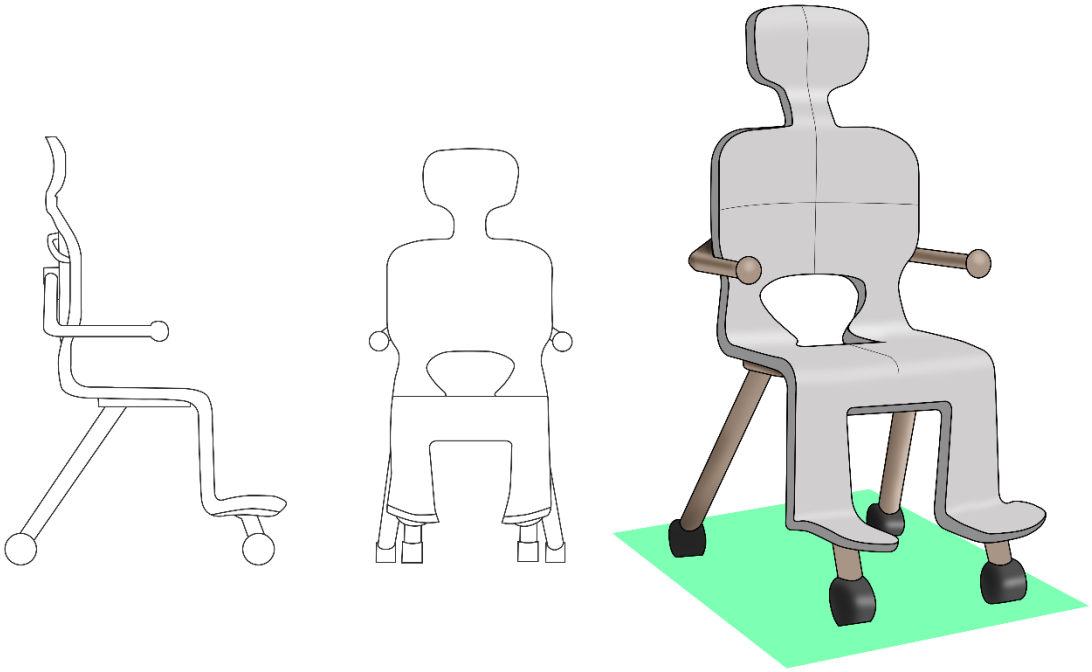


Figure 30: Sixth concept



Figure 31: AI contextual render of the sixth concept

Concept 6 (Figure 30) The sixth concept explores what the chair might look like if the entire main body were manufactured as a single large component, a form that loosely echoes the silhouette of a seated human figure, incorporating the headrest, footrests, and other elements as integrated details. While visually distinctive, this approach significantly limits adjustability and creates practical challenges around transportability. Etac's chairs are currently shipped pre-assembled, and a single-body design would make packaging considerably less efficient, increasing both transport costs and carbon emissions.

4.3.2 Concept Selection

With six concepts developed, a structured selection process was conducted to identify the most promising direction for further development. Involving external stakeholders in this process was important to minimize designer bias and broaden the basis for decision-making. The selection took the form of individual workshops in which each concept was presented to participants through sketches, renders, and physical mock-ups (Figure 33). Participants were then asked to rank all six concepts from 1 to 6, where 1 indicated the strongest perceived potential and 6 the weakest.

The participants were drawn from different roles within Etac's research and development team, which meant their priorities naturally differed. For example, the product manager (Participant 6) placed greater emphasis on caregiver accessibility, while the design engineer (Participant 2) focused more heavily on structural integrity and manufacturability. The individual rankings were compiled into a combined score

matrix (Figure 32), and the concept with the lowest total score was identified as the preferred direction.

Attende	C1	C2	C3	C4	C5	C6
1 (Design engineer)	3	5	2	4	1	6
2 (Design engineer)	2	6	1	3	4	5
3 (Design engineer)	2	5	1	3	3	6
4 (Development manager)	2	3	1	5	4	6
5 (Design engineer)	1	2	5	4	3	6
6 (Product manager)	1	5	4	6	2	3
	11	26	14	25	17	32

Figure 32: Concept selection matrix

Concept 1 achieved the lowest combined score of 11, making it the selected concept for further development. It should be noted that Concept 1 was not in its final form at this stage, it continued to be refined throughout the subsequent development process, incorporating elements from other concepts where appropriate.

4.3.3 Mock-ups

Physical mock-ups were produced continuously throughout the project to evaluate individual components and functional details of the mobile shower commode chair. 3D printing was the preferred fabrication method, as it allows models to be produced passively and used directly for testing and evaluation without significant post-processing.

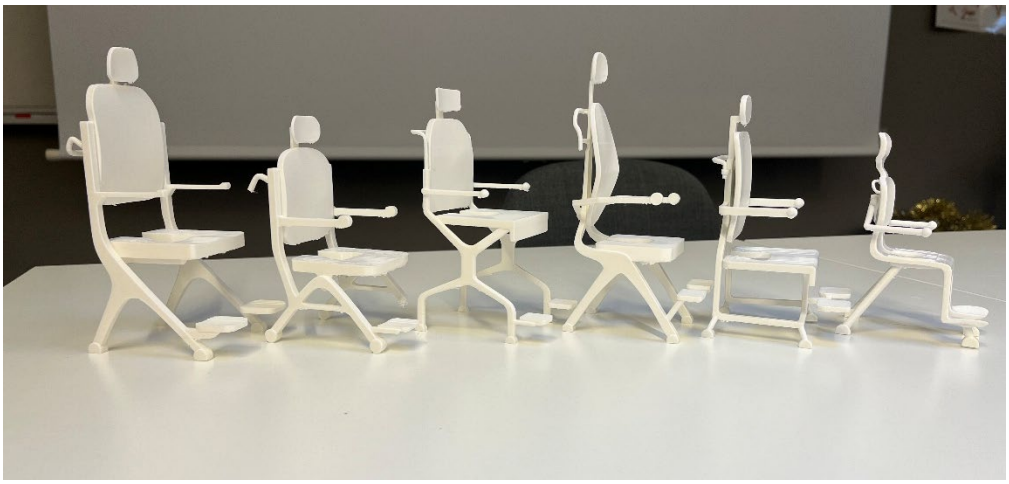


Figure 33: 3D-printed scale models of the six concepts used in the concept selection

To support the concept selection workshops, simplified scale models of all six concepts were produced (Figure 33). The concepts were first modelled in Autodesk Alias at a level of detail sufficient for proportional evaluation, a process that took hours rather than the days that fully resolved models would have required, and then 3D-printed at a reduced scale. These models gave stakeholders a physical point of reference that sketches and renders alone could not provide.

4.3.4 Design features and iterations

The following sections describe the development of the individual design components of the mobile shower commode chair: the armrests, adjustment mechanisms, footrests, neck rest, backrest, seat, and frame. Each component went through multiple iterations before reaching its final form. Relevant ergonomic considerations are also described for each component.

Ergonomic dimensions throughout the design process were based on Etac's existing product guidelines and anthropometric data extracted from the software PeopleSize. The relevant measurements are provided in the Appendix.

Armrest

Several mock-ups of the armrest were produced at full scale to evaluate shape and proportion in a physical context (Figure 34). The iterations can broadly be divided into two groups based on the form of the grip end: ball-ended and cylindrical.



Figure 34: 3D-printed mock-ups of different armrests

The ball-ended version was explored because the shape affords a more flexible grip, the user can approach it from different angles, which is particularly useful when rising from the chair. The relatively narrow width also makes it easy to grasp. The cylindrical version, by contrast, was more comfortable during passive arm resting but offered less flexibility. Various profiles for the main body of the armrest were also evaluated, including concave, convex, and a "wing" shape. The wing variant was discarded early in the process, as it locked the arm into a fixed position, an approach incompatible with a product designed for a highly varied user group.

The ball-ended version was ultimately also discarded, as testing revealed that it did not sit naturally in the hand. The cylindrical form felt more intuitive in a neutral resting position. Comfort was prioritized over flexibility in this decision. A flat, concave profile was selected for the main body, as a convex surface would constrain the arm angle in a way that may be uncomfortable for some users.

The final armrest length is based on the mean back-of-elbow to grip measurement for German males and females aged 65 and over, approximately 326.8 mm. The width is 40 mm, sufficient for resting the arm while also allowing a secure grip. Armrest height is initially at 221 mm above the seat surface, based on the mean forearm height for the same demographic group.

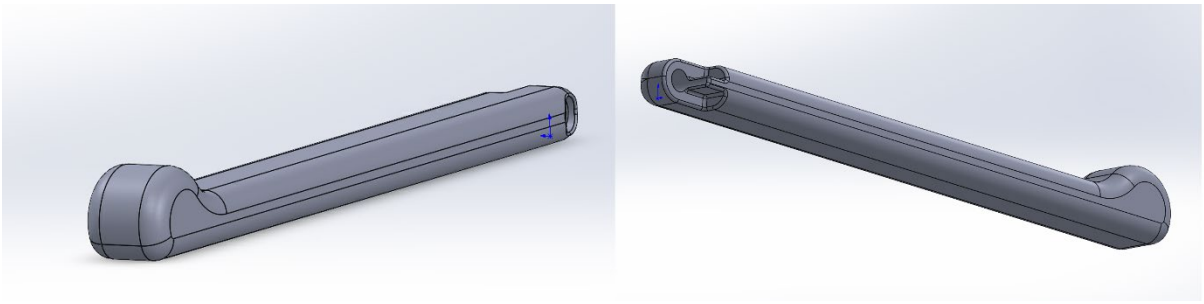


Figure 35: Final armrest design

The armrests must also be foldable. A non-foldable armrest would prevent the primary user from accessing the chair from the side, important when transferring from a bed, for example. The armrest connects to a mechanical joint in the backrest (Figure 36) with two defined positions: one at 90° horizontally and one at slightly more than 90° vertically, reducing the likelihood of the armrest falling back unintentionally when left in the raised position.

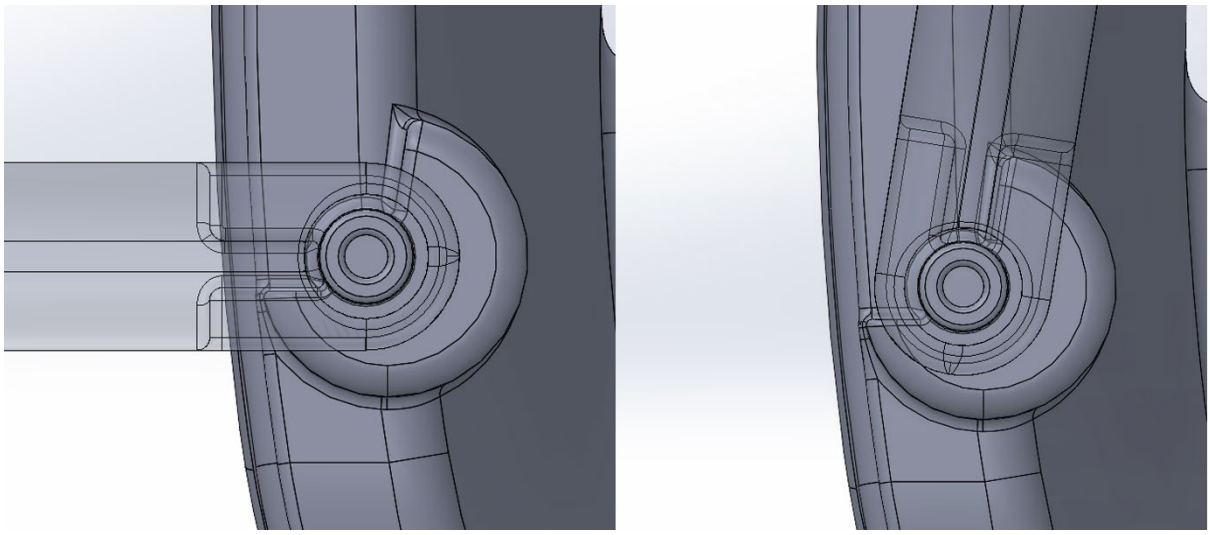


Figure 36: Folding mechanism of the armrest

Vertical adjustment mechanism

One concrete way to improve the intuitiveness of a product is to simplify the mechanisms users must interact with. The vertical adjustment mechanism (Figure 38) operates through a single button press combined with a supporting hand on the component being adjusted. Pressing the button releases an internal bracket inside the tube; the connected component can then move freely in either direction until the user releases the button, at which point a spring pushes the bracket into the nearest locking slot, securing the position.

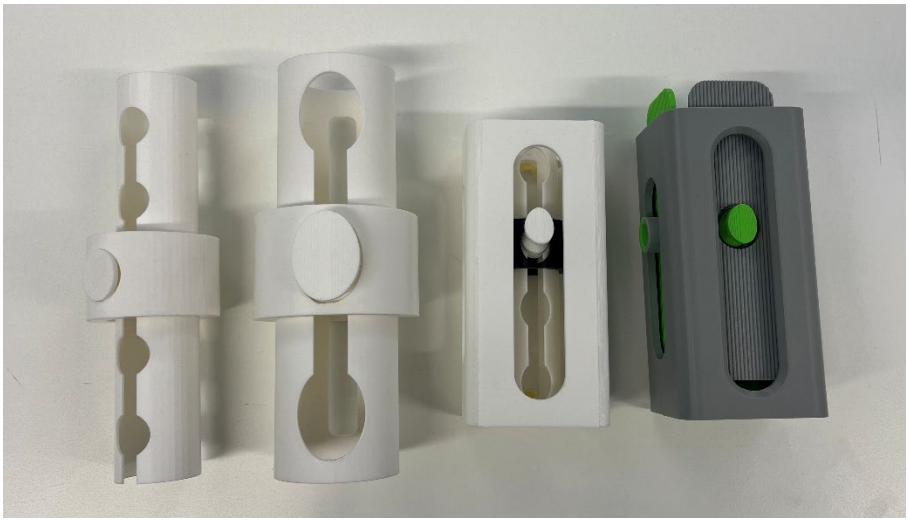


Figure 37: 3D-printed mock-ups of an adjustment mechanism

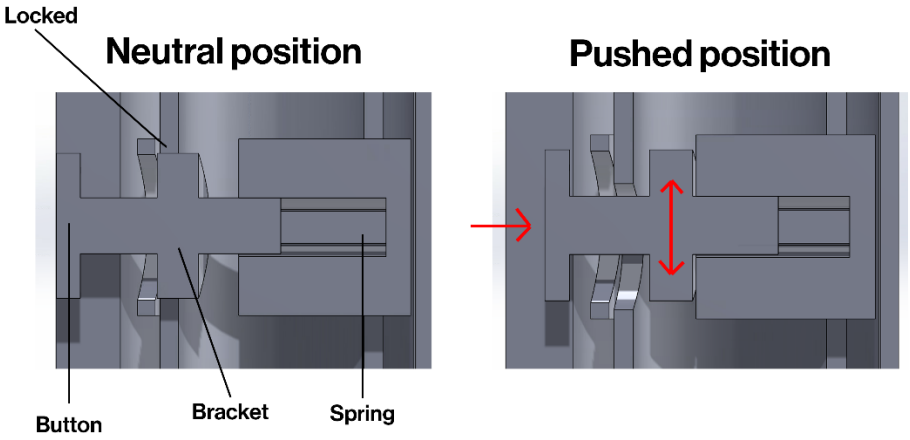


Figure 38: Vertical adjustment mechanism

Several variations of the mechanism were developed and tested (Figure 37). The initial version was designed for the footrest attachment and featured a circular cross-section to match the stainless-steel pipe. A rectangular version was subsequently developed for the backrest, functioning identically but adapted to a different profile geometry (Figure 39). Keeping the mechanism as consistent as possible across all adjustable components was a deliberate design decision, aimed at reducing the cognitive load placed on caregivers, understanding one mechanism means understanding all of them.

The mechanism remained largely stable throughout the process; it performed reliably from the outset. The primary area of iteration was dimensional refinement, with various sizes tested to achieve the smallest possible footprint without compromising function. Pipes smaller than Ø40 mm introduced tolerance issues, most notably the button becoming prone to jamming, a problem that would likely be resolved in a production context using more precise materials and manufacturing processes.

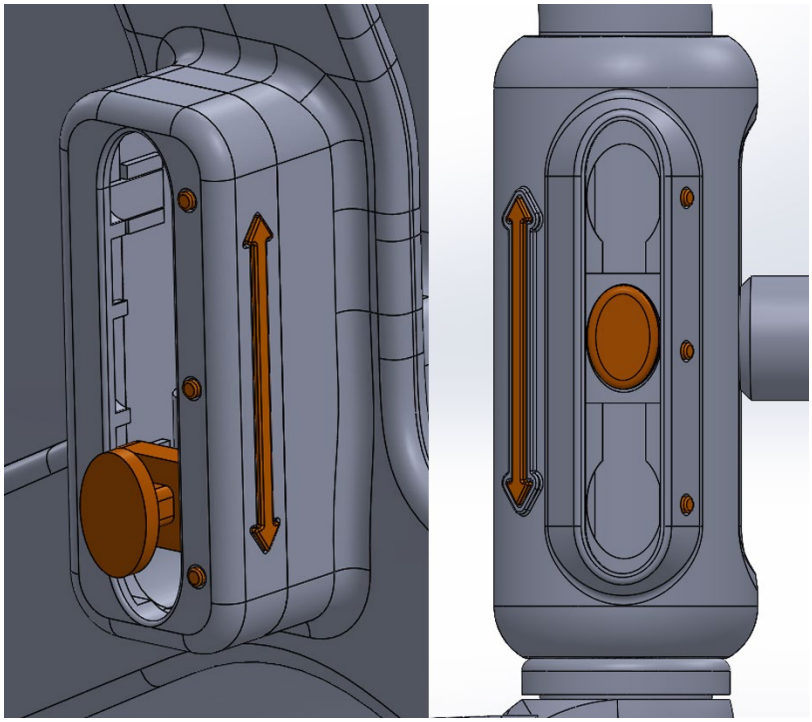


Figure 39: Rectangular and circular vertical adjustment mechanisms

To support visual communication of the mechanism's function, arrows were added adjacent to each adjustment point to indicate the direction of movement, and three dots were added to indicate the available locking positions.

Footrest

While the footrest may appear to be one of the simpler components of the mobile shower commode chair, it plays an important ergonomic role, stabilizing the user's feet and maintaining the 90-degree angles at both the knee and ankle that are essential for correct seated posture.

Since the chair is used in wet environments, slip resistance is an important consideration. A curved surface profile was introduced to the footrest to encourage water runoff and prevent pooling (Figure 40). This curvature also adds visual complexity to what would otherwise be a flat surface, contributing positively to the overall aesthetic character of the chair and aligning with the "Midimalism" trend identified in the design research, functional, but visually refined enough to integrate into a contemporary bathroom environment.

Footrest dimensions are based on the mean foot length of German males and females aged 65 and over, which is approximately 245.5 mm. The width is a minimum of 95.3 mm, based on the mean foot width for Belgian males and females aged 65 to 80, ensuring the footrest covers the full width of the foot.

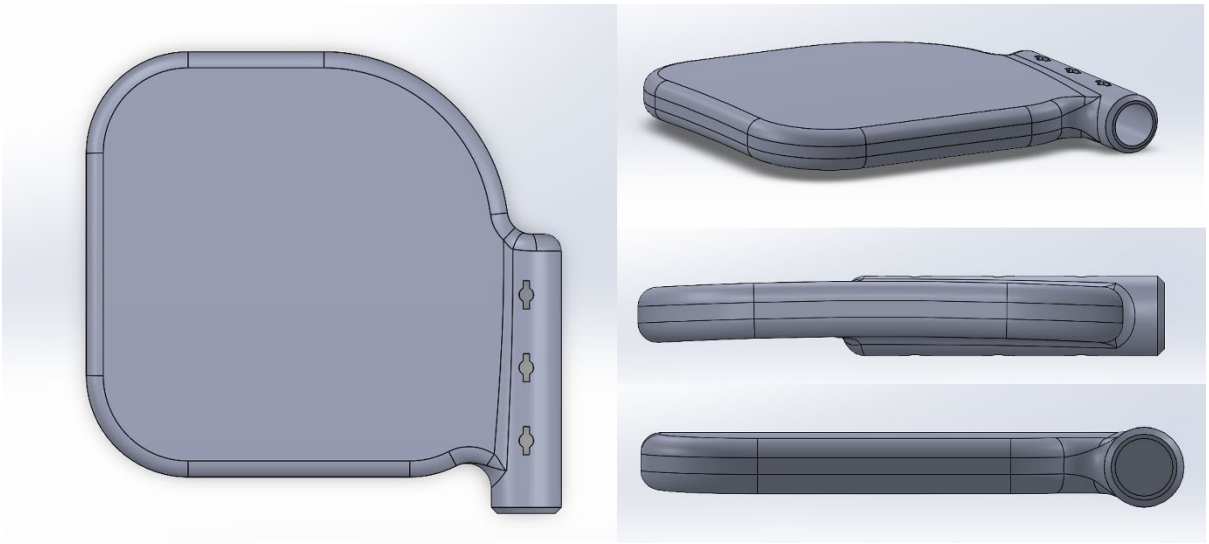


Figure 40: Footrest of the mobile shower commode chair

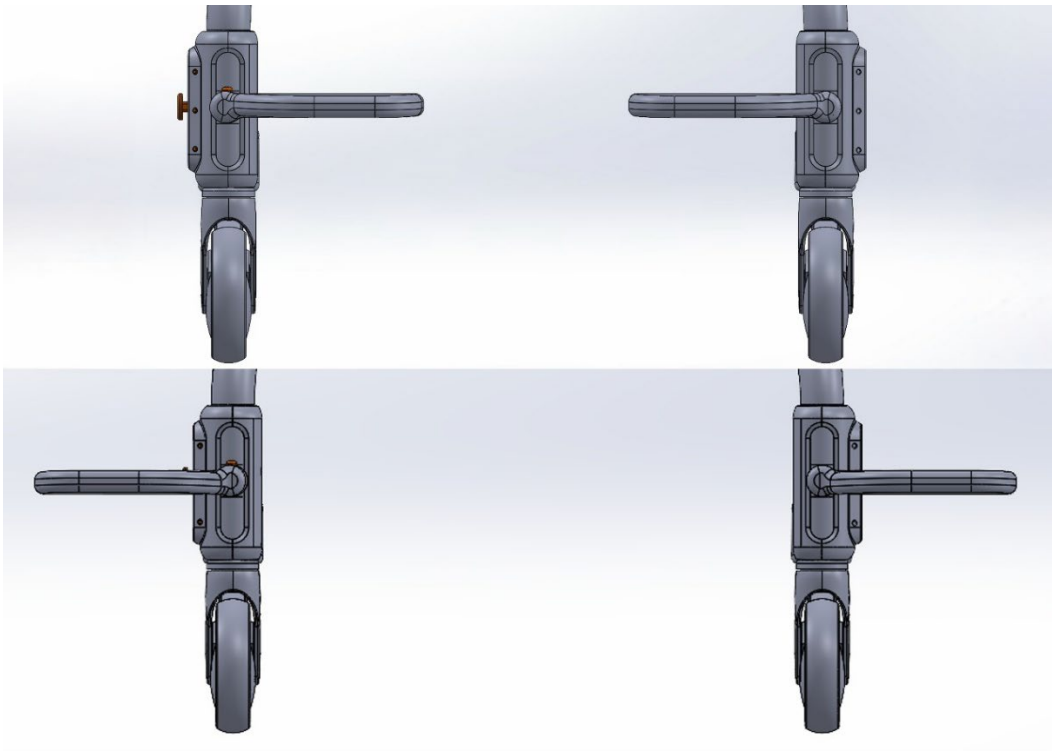


Figure 41: Footrests from original position to 180° rotation

The footrests must be removable or repositionable, as some users are unable to mount the chair with them in the forward position. The design addresses this through a symmetrical mounting tube that can be rotated 180°, swinging the footrests out of the transfer zone (Figure 41). Height adjustment is achieved through a pin mechanism with three defined positions (Figure 42). The full adjustment mechanism used elsewhere on the chair was considered for this application but was rejected, the footrests are light enough that the spring-loaded bracket mechanism would be unnecessarily complex and would add visual bulk to what is intended to be a slender component.

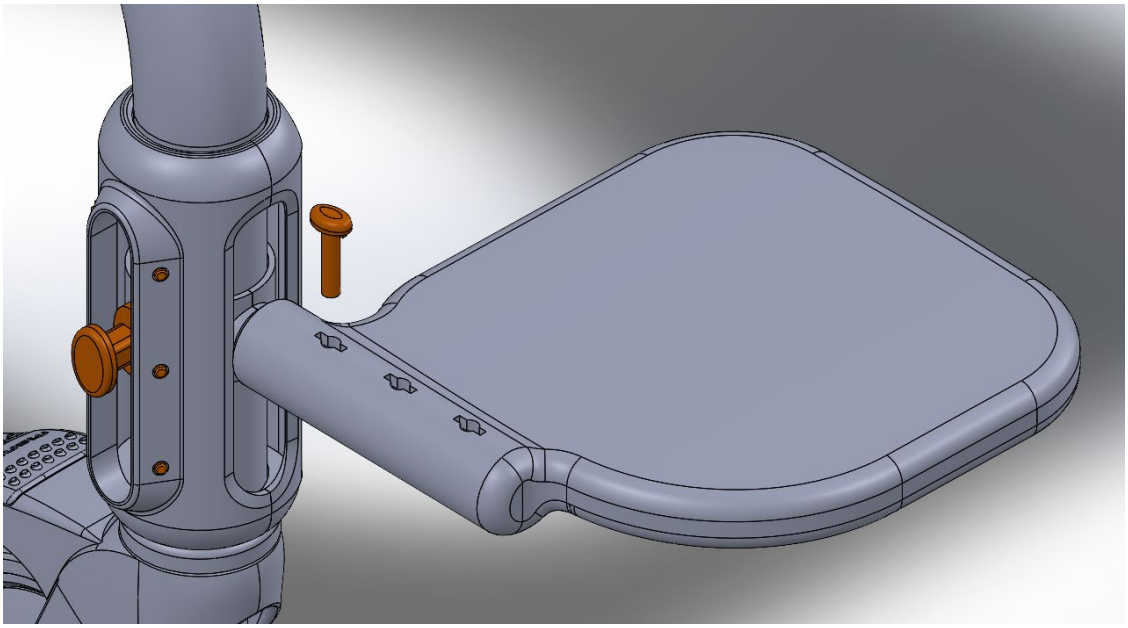


Figure 42: Footrest with pin and vertical adjustment mechanism

Neck rest

Many users experience weakened shoulder and neck musculature and tend to lose upright posture during extended periods of seated use. Adequate neck support is therefore important to reduce strain, though the neck rest for this application should not resemble a conventional headrest of the kind found on office chairs or car seats, which are designed for reclining positions, but rather a support that functions in an upright seated posture.

The minimum width of the neck rest is based on the mean neck width just below the Adam's apple for German males and females aged 40 to 65, which is approximately 130 mm. The neck rest also follows the curvature of the head and neck to provide lateral support when the user's head tilts. The relevant measurement for this curvature, from the back of the head to the broadest point of the cheekbones, is approximately 123.5 mm for German males and females aged 65 and over. The overall height of the neck rest follows Etac's current neck rest specification of approximately 140 mm.



Figure 43: Neck rest with adjustments

The neck rest is the most adjustable component on the chair, with provisions for both height and angle adjustment to accommodate the range of user proportions (Figure 43). An initial plan to use the same vertical adjustment mechanism as the backrest and footrests was abandoned after mock-up testing, the mechanism created an unacceptably bulky profile at the neck rest. A friction-based vertical adjustment was developed instead, in which a plate connected to the neck rest sits within a tight channel, held in position by friction and released with deliberate force (Figure 44). This approach is familiar from office chair headrests, which was seen as an advantage since users who recognize the mechanism from a domestic or workplace context may be more comfortable with it in their home, helping to reduce the clinical associations of the product.

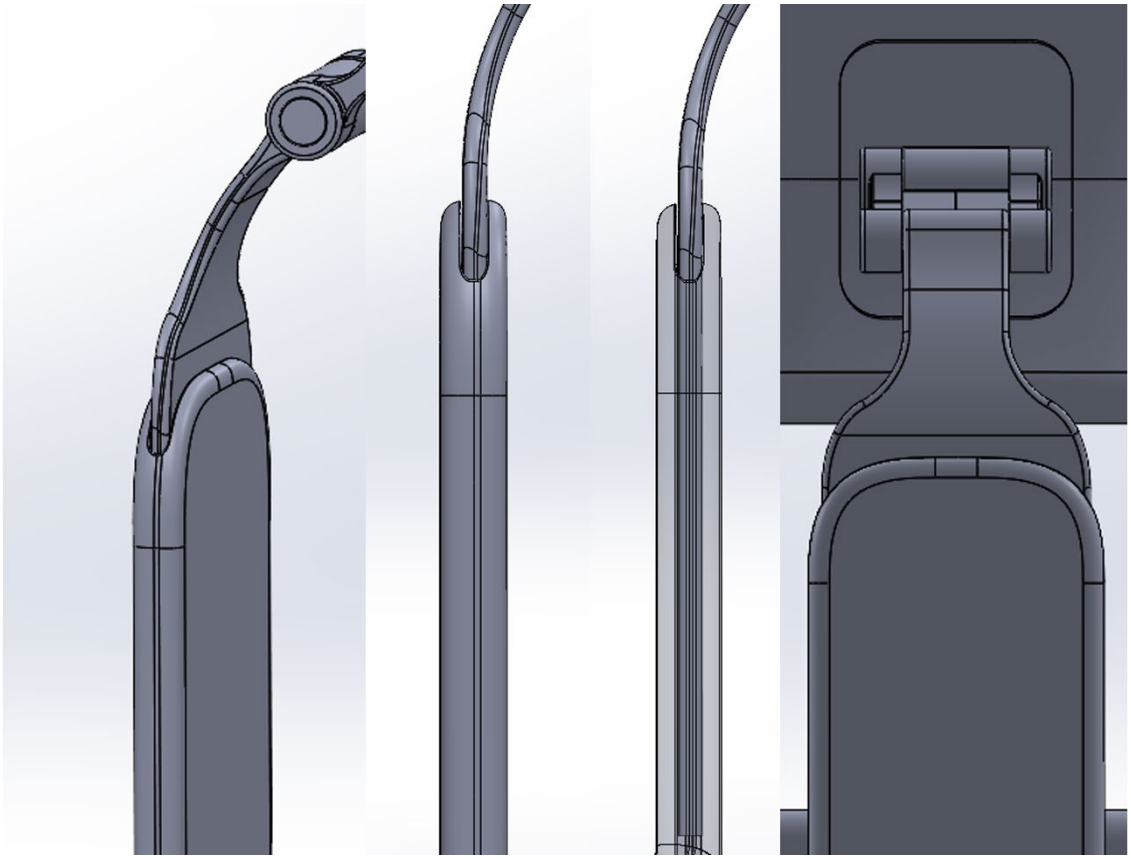


Figure 44: Neck rest vertical adjuster

Backrest

The primary ergonomic function of the backrest in a mobile shower commode chair is to support a neutral seated posture, maintaining a 90-degree angle between the torso and hips and supporting the natural lumbar lordosis of the spine. The backrest should therefore not be fully vertical. Additionally, since many users struggle to maintain an upright position independently, a degree of lateral torso support is beneficial.

The backrest cannot extend to cover the user's full back, as the caregiver requires access from behind the chair beneath the backrest. The access gap must be a minimum of 150 mm. The seated shoulder height for German males and females aged 65 and over is approximately 570.5 mm (Acromion height, sitting), which informs the upper boundary of the backrest. As PeopleSize did not provide back width data, the backrest width follows Etac's recommended dimension of approximately 420 mm, resulting in an overall backrest size of approximately 420 × 420 mm. The backrest is adjustable across three dimensions to accommodate variation in user stature.

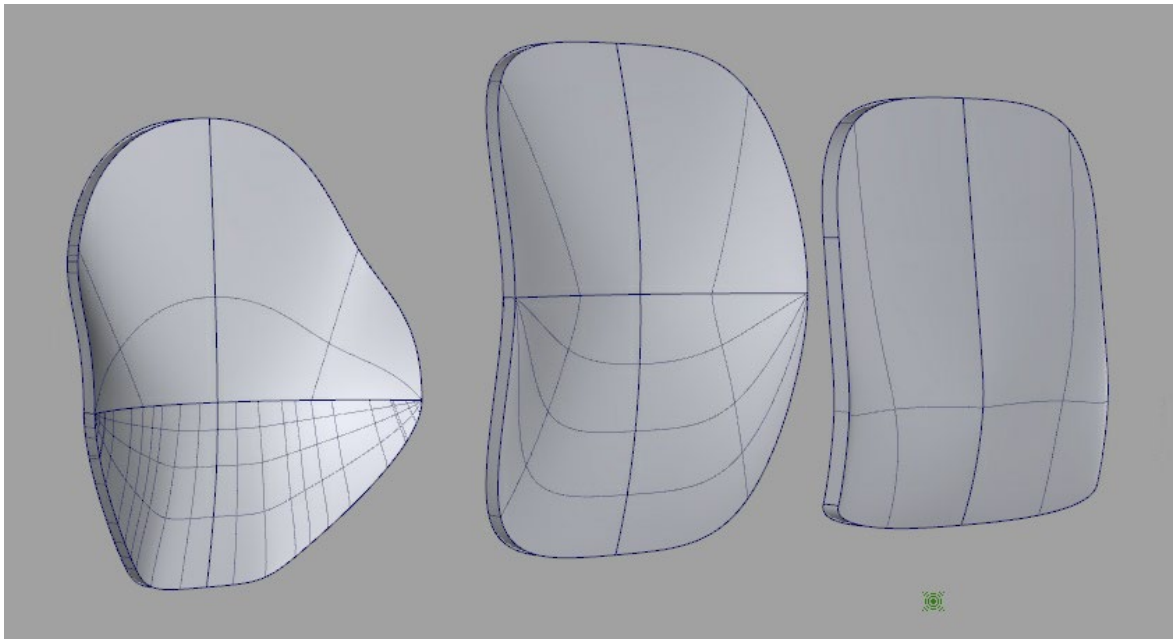


Figure 45: Three initial versions of the backrest

Several backrest forms were explored during development (Figure 45). Incorporating lateral torso support was desirable given user feedback that maintaining upright posture on current chairs is difficult. However, widening the sides of the backrest introduces practical constraints, it reduces the space available for armrests and increases the gap between the armrests and the user's body, reducing the effectiveness of the armrests as a stabilizing surface. The final backrest (Figure 46) draws on the geometry of a hyperbolic paraboloid, a saddle surface that is simultaneously convex and concave. This form supports neutral lumbar lordosis while also providing a gentle degree of lateral torso support. The inherently organic nature of the hyperbolic form also aligns well with the biophilic design trend identified in the research.

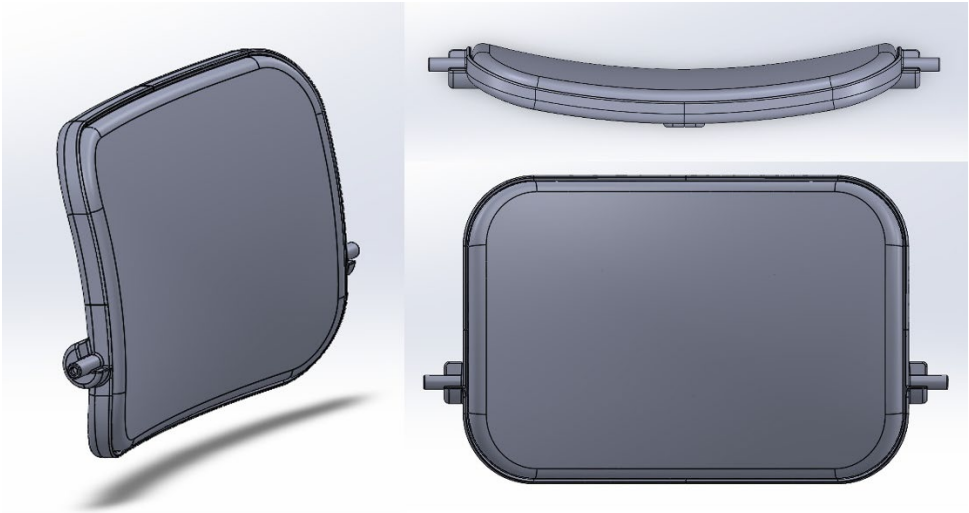


Figure 46: Final backrest

The backrest is constructed in two parts (Figure 47): a thin polypropylene shell, which provides structural rigidity and serves as the mounting interface with the frame and armrests; and a thicker polyurethane foam pad, which forms the contact surface for the user's back. The foam is soft, easy to wipe clean, and chemically resistant which is well suited to the demands of the application.

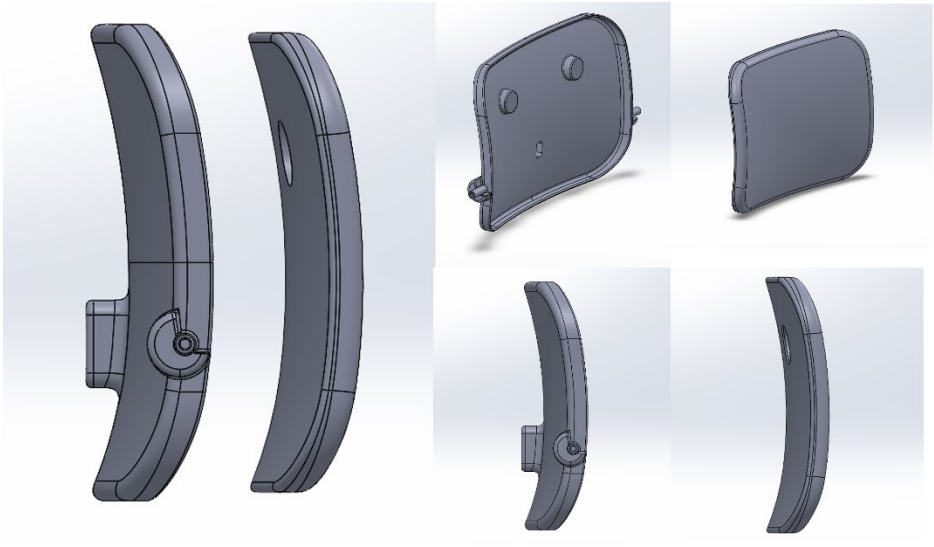


Figure 47: The two different parts of the backrest

Seat

A well-designed seat promotes even pressure distribution, reducing the risk of pressure ulcers, while also facilitating the correct 90-degree angles at both the hip and the knee. To accommodate the diversity of the user group, the seat is designed to be as neutral in form as possible.

The seat width is based on the mean hip breadth for Belgian males and females aged 65 to 80, which is approximately 395.8 mm. The seat depth does not exceed 483.5 mm, the mean buttock-to-back-of-knee (popliteal tendon) measurement for German males and females aged 65 and over, to ensure the 90-degree angle at the knee is not compromised.

As with all mobile shower commode chairs, a cutout is incorporated into the seat to enable toileting while seated. The cutout is 190 mm wide and 400 mm in length, dimensioned to be large enough to avoid discomfort for smaller users while not compromising the structural integrity of the seat or causing difficulty during use (Figure 48).

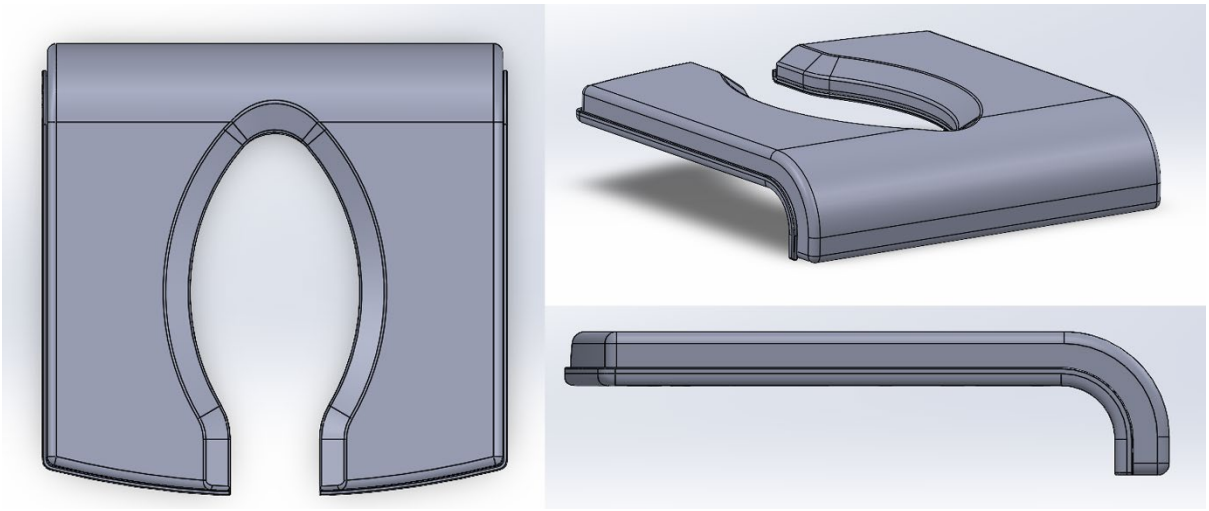


Figure 48: Final seat

The seat, like the backrest, is constructed in two parts: a rigid polypropylene base shell and a polyurethane foam pad. An additional consideration specific to the seat is water retention between the two components, which could compromise hygiene and material integrity over time. To address this, drainage holes were incorporated into the base shell to prevent water from accumulating between the layers (Figure 49).

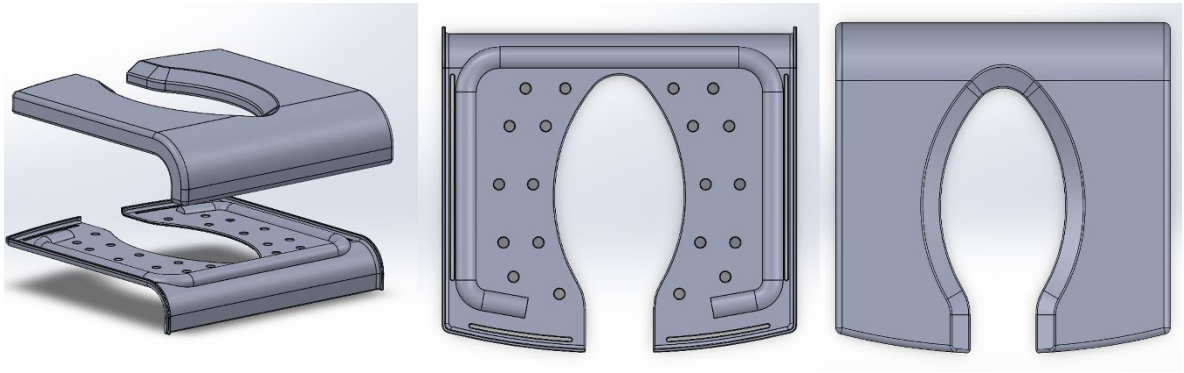


Figure 49: The two different parts of the seat

The overall profile of the seat pad is intentionally flat and neutral. Since the foam will conform to the user's body under load, a more pronounced sculptured form would risk fitting fewer users well without providing meaningful additional comfort.

Frame

The frame underwent the most extensive development of any component, evolving considerably from the version presented during the concept selection. The initial iteration consisted of two fully planar side profiles (Figure 50), envisioned as stainless-steel tube structures enclosed in a plastic cover to remove the exposed-tube aesthetic of current market products. Shortly after the concept selection, however, this approach was reconsidered. The plastic cover served a purely aesthetic function with no structural benefit, and as an outer surface it would be vulnerable to impact damage and

cracking over time. A version was also developed in which the two side profiles were connected into a single backbone, drawing from the language of Concept 4, but this direction was ultimately set aside as well (Figure 51).



Figure 50: First iteration of frame for the first concept

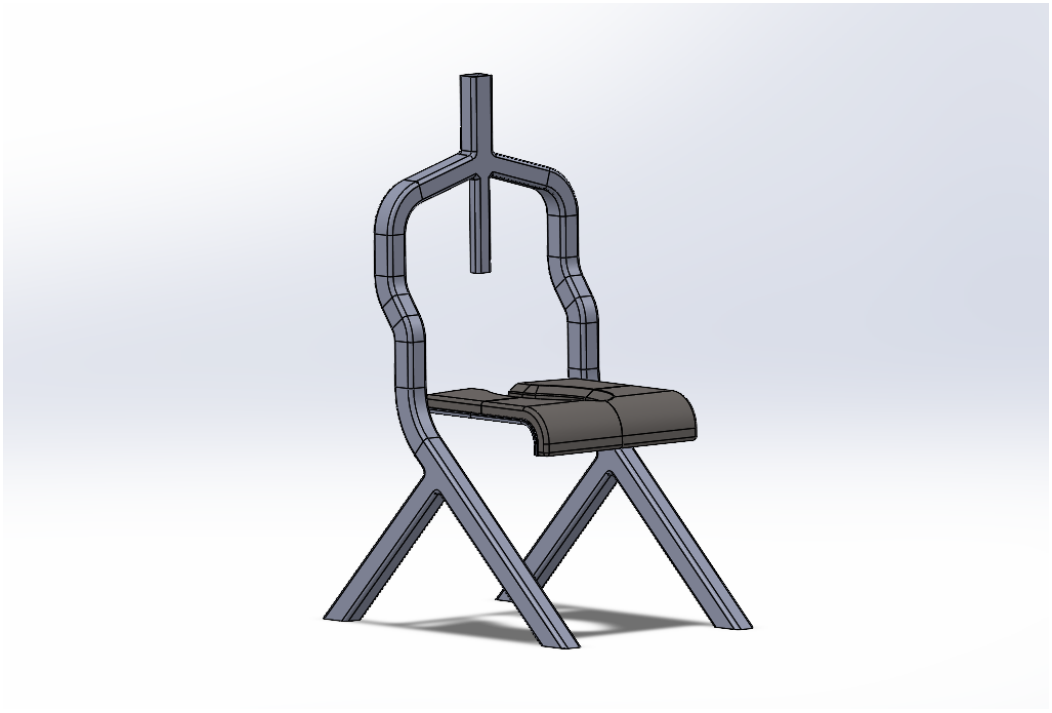


Figure 51: Early iteration of the first concept with a joint frame

Further iterations explored different framings of the box-shaped side profile before the approach was abandoned entirely in favor of a conventional stainless-steel tube structure, the same construction method used by most competitors. While this represents a more familiar approach, the goal remained to differentiate the chair

through the shaping and arrangement of the tubes rather than through a fundamentally different construction system (Figure 53).

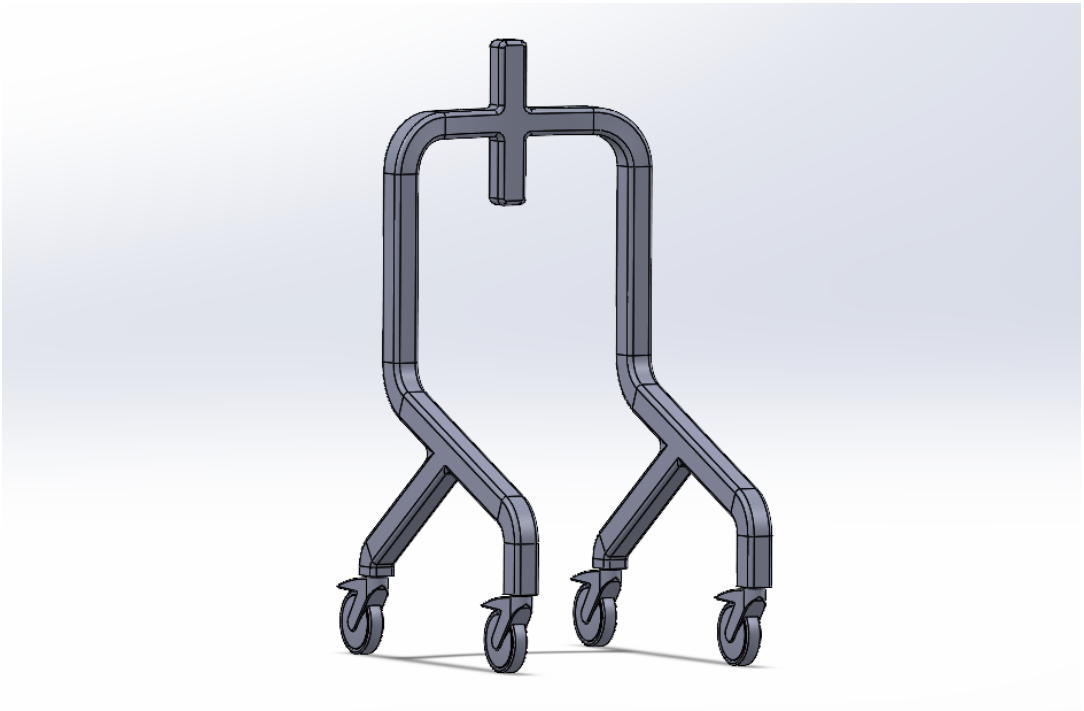


Figure 52: Another iteration of the first concept with more straight lines



Figure 53: First version of the first concept with the tube structure

The primary tube forms an inverted U-shape, defining the upper frame and caregiver grip area. Two separate rear leg tubes are welded onto the main frame, with a cover component concealing the welds while also serving as a transition from frame to seat. Multiple iterations of this cover were explored to find a form that read as a natural extension of the seat rather than an applied detail (Figure 54).



Figure 54: 2D sketch iterations of the casing to cover the welds

After several iterations in which the chair was felt to lack sufficient visual distinctiveness, a decision was made to introduce three-dimensionality to the leg geometry, inclining the legs inward to create a more organic, dynamic silhouette (Figure 55). This inclination aligns the chair's structural character with the biophilic design trend and gives it a visual identity that distinguishes it clearly from the angular, orthogonal forms of current products. A central connecting tube was also added at this stage, linking the rear legs through the weld cover and improving overall structural coherence.



Figure 55 First iteration of the chair with inclined legs

A transition from a 40 mm to a 25 mm tube diameter was introduced at the top of the frame where the caregiver grips are located (Figure). The larger diameter provides structural rigidity throughout the main frame, while the thinner upper section, combined with the grip covering, produces a more ergonomically appropriate handle diameter for the caregiver.

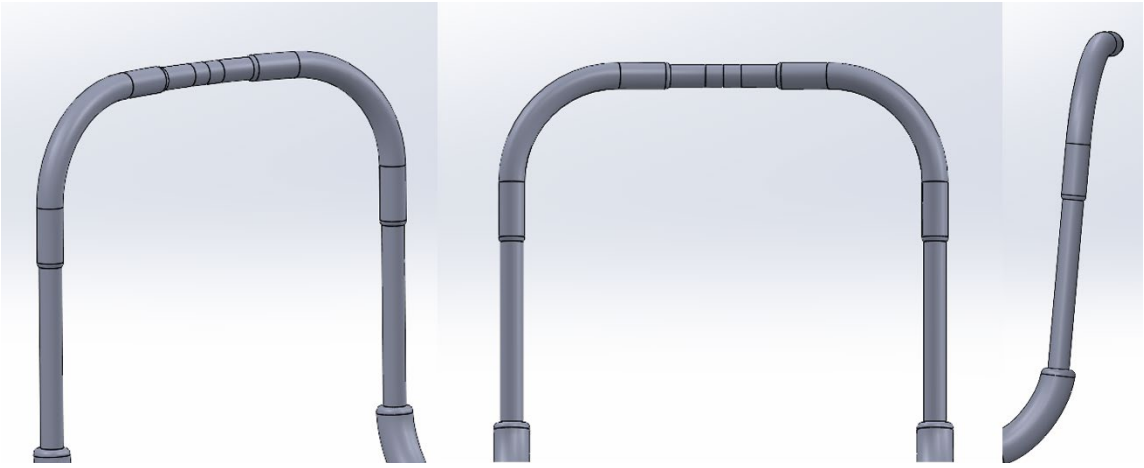


Figure 56: Grip for ergonomical handling of the commode chair

The grip is positioned on the curved section of the frame and extends slightly onto both the vertical and horizontal portions, giving the caregiver flexibility in hand position and the ability to adjust their grip during prolonged use (Figure 56).



Figure 57: Backbone of the mobile shower commode chair

To connect the backrest, neck rest, and armrests to the frame in a cohesive way, a backbone component was designed (Figure 57). Taking inspiration from both other concepts and the biophilic design direction, the backbone features a flat, flowing form with a gentle swerving profile, a visual quality rarely found in clinical products. The backbone attaches to the frame at three snap-fit points positioned around the grip section. The inward inclination of the upper frame geometry means that the backbone cannot be dislodged by incidental contact. The backbone also creates a coherent visual relationship between the backrest and the neck rest, an integration that is notably absent in many current designs, where the neck rest appears as an afterthought rather than an intentional part of the design system.

4.4 Delivery

This section describes the outcomes of the methods used to present the final concept in response to the project's research questions.

4.4.1 Final concept compliment with product specifications

The product specifications were compiled into a comprehensive table. The final concept does not satisfy all 51 specifications, as certain compromises were necessary to prioritize other requirements or accommodate constraints inherent to the project's scope. A full overview of which specifications the final concept complies with, could comply with, and will not comply with is provided in the Appendix.

Two specifications were assigned a "Will not comply" status. The first is M03: "The chair allows unified control of wheel locking." A centralized locking system would require either an electronic solution or a significantly different lower frame configuration designed specifically to accommodate such a mechanism. While this would not fundamentally undermine the design, it would alter the lower frame geometry in ways that compromise the aesthetic intent of the chair. This specification was therefore set aside in favor of visual cohesion.

The second is E04: "The chair provides adjustable steering handles." Adjustable handles were excluded on the grounds of intuitiveness. Introducing additional adjustment points to the caregiver grip area would increase the complexity of operating the chair and risk adding confusion for caregivers who are already managing multiple adjustment functions. The time cost to the caregiver of locating and operating additional adjustments was considered to outweigh the benefit.

4.4.2 3D Physical model

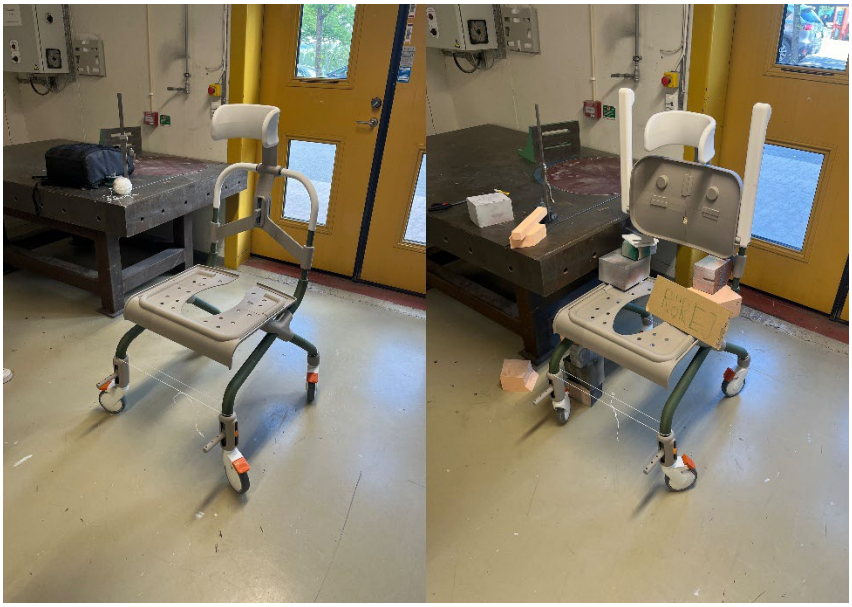


Figure 58: 1:1 3D Physical model

A 1:1 scale physical model was produced to present the final concept in a fully interactive, three-dimensional form with functional adjustment mechanisms (Figure 58). The model comprised 94 individually 3D-printed parts (Figure 59). Larger components that required splitting for printing were subsequently spackled, sanded, and painted to remove the splitting lines necessary to fit the different parts onto the bed of the different 3D-printers used (Figure).



Figure 59: 3D-printed parts

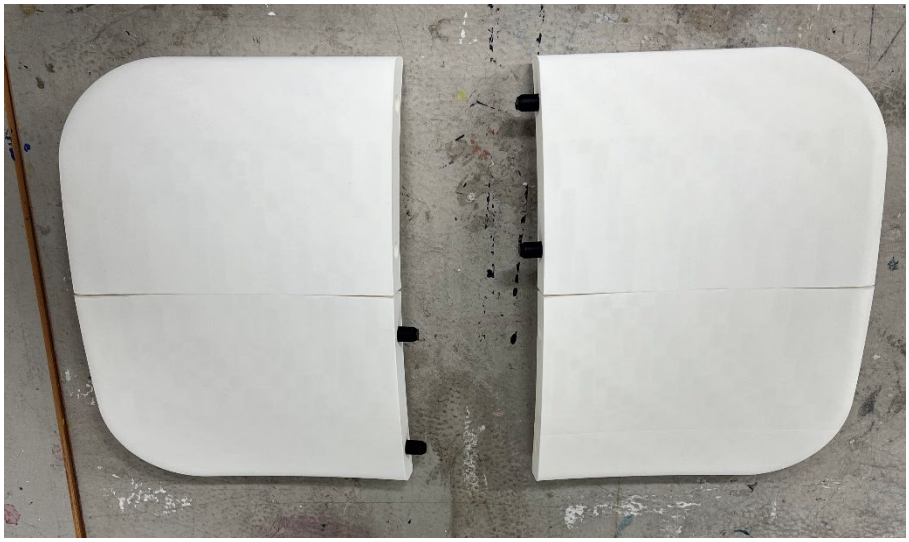


Figure 60: Raw look at the backrest

To manage the complexity of producing 94 parts, a tracking table was maintained throughout the production process (Figure), recording each part, its target color, and its print status. In the "Printed?" column, green indicates a completed part, yellow indicates a part currently in progress, and red indicates a part not yet printed

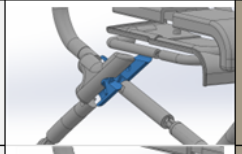
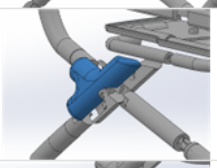
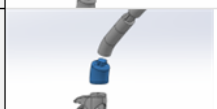
Part	Image	Color	Printed?
1			
2			
3			
4			
5			
6			
7			

Figure 61: Prototype-parts checklist

5 Final model



Figure 62: Studio photo render of the final concept

The color palette mentioned in chapter 4.2.3 was used for the final concept. The whiter color (RAL 9010) was used for the areas in which physical contact will occur which includes the seat, backseat, neck rest, footrests, armrests, and the handles on the backside of the chair. RAL 1015 was used for smaller parts of the chair which would create a needed contrast with the other colors, see figure. RAL 6003 was used for the frame of the chair, the olive green creates a good contrast with the white and the darker tone grabs less attention, so that the functional parts of the chair are highlighted. Last, the orange (RAL 2004) where used for the interactive parts of the chair. The buttons for the vertical adjustment mechanism and the plug for the footrest horizontal adjustments are RAL 2004 to make them truly stick out and be as intuitive as possible for the caregiver to know where to adjust the different parts of the chair (figure 62).



Figure 63: Front views of the mobile shower commode chair

Figure 63 highlights the front part of the chair, especially with color contrast of the different parts and some of the details of the different front parts.



Figure 64: Back views of the mobile shower commode chair

Figure 64 shows the backside of the chair with more details such as the backbone and a better view of the overall frame.

6 Discussion and conclusion

This chapter discusses the results of the project in relation to the two research questions and reflects on the methods used. Limitations that affected the outcomes are also addressed. The chapter ends with conclusions and suggestions for further research.

6.1 Discussion of results – Research question 1

The first research question asked what the specific requirements are for developing a next-generation mobile shower commode chair for home-based, single-person healthcare. The pre-study gave a thorough answer to this question. By combining a STEEP trend analysis, a competitor analysis, user statements from Etac, and 51 product specifications, a clear picture of what the chair needs to achieve was established.

A few strong themes came out of this work. Healthcare is moving into the home, which means the chair needs to work well in domestic environments rather than clinical ones. This affects everything from how much space the chair can take up to how it looks in someone's bathroom. The rise of single-person households adds to this, since the chair needs to be operated by one caregiver alone without things becoming complicated or time-consuming. The trend analysis also showed that users are increasingly unwilling to have products in their home that look like hospital equipment, which is a gap that most current products on the market do not address at all.

One limitation worth mentioning is that no primary data was collected directly from caregivers or users. The user needs were based on secondary data from Etac, which is useful but still filtered through the company's perspective. Direct interviews or observations with caregivers working in real home environments would have given more details and unfiltered pictures of the requirements. It should also be mentioned that it is hard, according to Etac, to get hold of the caregivers or the primary user. Their busy schedule cannot prioritize a thesis student when Etac already has a large bank of valuable research. This is, though, still something to keep in mind when evaluating how complete the answer to the first research question really is.

6.2 Discussion of results – Research question 2

The second research questions asked how the identified requirements could be translated into the design of a next-generation mobile shower commode chair. The design process, which included brainstorming, concept selection, iterative development, and a 1:1 scale prototype, shows a structured way of working through this translation.

Several design decisions connect clearly back to specific requirements. The organic shapes of the frame came directly from the trend analysis finding that future bathrooms

will move towards biophilic, nature-inspired aesthetics. The orange accent color used on all adjustable parts was a direct response to the need for intuitive single-person operation, if the caregiver can spot one adjustment point, they can spot all of them. The saddle-shaped backrest addresses the ergonomic need to support the natural curve of the spine while also giving some lateral support to users who struggle to sit upright. The foldable armrests, rotating footrests, and consistent adjustment mechanism all respond to the need for efficient, low-step operation by a single person.

With that said, not everything went according to plan. Two specifications were not met in the final design, centralized wheel locking and adjustable steering handles. Centralized wheel locking was left out because implementing it mechanically would have required a very different lower frame structure, which would have compromised the look of the chair. It is a reasonable compromise within the scope of this project, but it is a real functional trade-off that would need to be solved in a proper production development phase. Adjustable handles were left out because adding more adjustment points to the caregiver grip area felt like it would create more confusion than it solved.

The lack of real user testing is also a significant limitation. Many specifications sit in the “could comply” category simply because there was no way to test them within this project. The prototype is convincing as a “looks-like” model and the adjustment mechanisms work, but it was never load tested, disinfection tested or evaluated by real users. It shows what the chair could be rather than proving that it works in every respect.

6.3 Discussion of methods

The Double Diamond worked well as an overall framework for the project. It gave the process a clear structure and made it easy to communicate what phase the work was in at any given point. In practice though, the process was more back-and-forth than the framework suggests. There were several moments during the development phase where earlier define-phase thinking had to be revisited, particularly when refining the frame and the backrest of the final concept. This is normal in design work, and the Double Diamond is flexible enough to handle it, but it is worth mentioning that the process was more cyclical than the chapter structure of this thesis implies.

The concept selection worked well. Involving people from different roles at Etac, design engineers, development managers, and a product manager which meant that the concepts were evaluated from multiple perspectives rather than just one. The main limitation is that all participants were Etac employees. Having caregivers or occupational therapists involved in the selection would have added a more direct user perspective to the decision.

The anthropometric data from PeopleSize was practical and appropriate for this project. The main limitation is that most of the data came from German and Belgian population

aged 65 and over, which may not perfectly represent users in Sweden or other markets. Including Swedish or general Scandinavian anthropometric data would have strengthened the dimensional decisions made throughout the design process.

The trend analysis was strong in most areas. The social, technological, ecological, and political. The economic section was weaker due to the broader aspects of the trends included. Many of the trends are hard to translate into product specifications or specific design choices. The STEEP method is good for covering a broader range of trends but in a project like this it only scratches the surface on how it impacts the product. More research on materials, production, etc., would have to be done to fully understand many of the trends implemented.

The AI image generation tool Nanobanana was used to produce contextual renders of the different concepts during brainstorming. This was a quick and effective way to show how concept might look in a real bathroom environment, and it was useful for communicating ideas to stakeholders. It should be noted though that AI renders cannot be read as precise design documentation but a solid way of visualizing a design.

6.4 Conclusion

This project set out to design a next-generation mobile shower commode chair for Etac, intended for home-based, single-person care in 10 to 15 years' time. The two research questions guided the work from start to finish, and the following conclusions are drawn in relation to each.

The pre-study successfully identified a broad set of requirements across functional, ergonomic, aesthetic, and regulatory areas. The most important finding was that the combination of home-based care, single-person households, and changing users' expectations create a design brief that current products on the market do not adequately meet. These requirements were structured into 51 product specifications that gave the design work a clear and traceable foundation.

The design process showed that these requirements can be translated into a coherent product concept. The final concept, with its organic frame, consistent color system for adjustable parts, ergonomically informed dimensions, and domestic color palette, is a genuine response to what was found in the pre-study. The 1:1 3D physical model gives Etac something tangible to work for in a future development.

It is important to be honest that compromises were made and that several specifications remain unverified. The final design is a directional proposal and a blueprint for future development rather than a finished product. Within the scope of a master thesis project however, the work delivers what it set out to, a well-grounded starting point for Etac's next generation of mobile shower commode chairs.

6.5 Suggestions for further research

The most important next step would be user testing with real caregivers and primary users in actual home environments. This would allow the design decisions made in this project to be properly evaluated and would likely surface refinements that could not be identified through the design process alone. It would also allow the many specifications currently in the "could comply" category to be assessed.

Structural and material testing would also be needed before the design could move towards production. Load testing, disinfection resistance testing, and waterproofing verification are all necessary steps that fall outside the scope of this project. The two specifications that were not met are both worth revisiting in a dedicated engineering development phase. And of course, a thorough DFM redesign or the chair to make it fully feasible.

Finally, the role of technology in future mobile shower commode chairs requires deeper exploration. The trend analysis identified AI companionship tools, smart living aids, and sensor-based monitoring as relevant emerging developments, and how these could be meaningfully integrated into a home-care product is an open and interesting design question for future research.

6.6 Personal reflection

This project has brought a lot of fun for me this last semester of the master program. Working on furniture and in the assistive products department has been a challenge to bring in certain elements while still justifying them accordingly to the research done. Focusing on designing a whole chair has been very challenging. To thoroughly work on every part to the best of ability has not always gone as I thought throughout this process and there are some parts which I am not truly satisfied with. If I went back in time to remake this project I would not focus on the whole chair. By putting all the energy on one part of the chair or creating a new mechanism such as the central locking system, all design elements would have been covered such as manufacturing, materials and creating an aesthetically pleasing shape. But I am still very happy with the results and the feedback which I have received on the project, and it has given me a great deal of experience. I would also like to thank everyone that has been involved in the project one way or another. From all the people in the design studio at Jönköping University and supervisor Peter for fun weekly meetings and discussions

Lastly, I would like to thank Etac for giving me the opportunity to do this project. Especially Mattias, Josefine and Maria who helped me with guidance throughout this project and everything related to mobile shower commode chairs and assistive products in general.

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8 Appendix

8.1 Appendix 1- Product specification list

User need	Metric	Value	MoSCoW	Final Concepts	
The chair complies with relevant safety and performance standards	Compliance with ISO 17966	Full compliance verified through certification	Must have		Will comply
The chair uses materials safe for skin contact	Compliance with ISO 10993	Full compliance verified through certification	Must have		Could comply
The chair withstands exposure to water	Waterproof rating	IPX6 or higher	Must have		Will not comply
The chair supports a wide range of users safely	Maximum safe working load	≥ 160 kg	Should have		
The chair complies with emerging sustainability standards	Compliance with ISO 14001	Full compliance verified through certification	Should have		
The chair complies with emerging material standards	Compliance with ISO 10993	Full compliance verified through certification	Should have		
The chair complies with emerging production standards	Compliance with ISO 9001	Full compliance verified through certification	Should have		
The chair can be maintained using components that remain available during supply chain instability	% of components with dual suppliers	≥ 90%	Could have		
The chair is be built from materials that remain cost-stable and widely accesible	Material price volatility	≤ ±10% over 12 months	Could have		
The chair avoids materials that are vulnerable to geopolitical shortages	Count of restriced materials	0 materials on EU Critical Raw Materials list	Should have		
The chair minimises environmental impact through material and energy choices	Recycled material content	≥ 50%	Should have		
The chair offers long term durability	Expected product lifespan	≥ 10 years under normal use	Should have		
The chair can be repaired, upgraded, or recycled to extend its lifespan	Number of replaceable parts	≥ 6 parts	Could have		
The chair enables smooth multidirectional movement	Multidirectional manoeuvrability radius	360° rotation with ≤ 20 N push force	Should have		
The chair provides secure wheel locking in all positions	Wheel lock engagement angle	360° lockability	Must have		
The chair allows unified control of wheel locking	Number of wheels controlled by central lock	All wheels (4/4)	Should have		
The chair rotates safely within a defined space	Rotation diameter	≤ 990 mm required clearance	Must have		
The chair maintains wheel stability during operation	Later stability angle	≥ 10°	Should have		
The chair compensates for reduced strength, balance, and mobility	Required operating force	≤ 20 N for all controls	Should have		
The chair supports hygiene toileting functionality	Opening dimensions	Meets ISO 17966 recommended aperture size	Must have		
The chair provides hygienic, wipe friendly surfaces	Cleaning time	≤ 2 minutes for full wipe down	Must have		
The chair withstands repeated chemical disinfection	Number of disinfection cycles without degradation	≥ 500 cycles	Must have		
The chair tolerates high temperature sanitation	Heat resistance tolerance	No deformation/discoloration at 85°C for 3 minutes	Should have		
The chair provides appropriate seating depth	Seat depth	465 mm	Should have		
The chair ensures ergonomic caregiver access	Backgrip to aperture distance	~540 mm	Should have		
The chair provides accessible transfer height	Seat height from floor	≥ 420 mm	Must have		
The chair functions effectively in small bathrooms without obstructing movement	Maximum footprint	≤ 930 x 650 mm	Must have		
The chair offers stable leg spacing for balance	Internal leg width	~400 mm	Should have		
The chair provides adaptable neck support	Neckrest adjustment range	≥ 80 mm vertical, ≥ 20° tilt	Should have		
The chair supports confortable arm positioning	Armrest height above seat	220-280 mm	Should have		
The chair accommodates different leg lengths	Footrest height adjustment range	340-465 mm	Should have		
The chair provides adjustable steering handles	Steering-handle djustment range	≥ 120 mm vertical, ≥ 30° angle	Could have		
The chair positions tilt-handle grips ergonomically with comfortable surfaces	Grip diameter and hardness	30-40mm; 20-40 Shore A	Should have		
The chair offers a seat made from soft, comfortable materials	Seat material hardness	15-35 Shore A	Could have		
The chair provides upper-body support	Upper-body support height	350-450 mm	Should have		
The chair incorporates ergonomic seating features	Pressure distribution variation	≤ 20% variation	Could have		
The chair includes armrests that minimise physical strain on arms and hands	Armrest height and width	200-250 mm length; 40-60 mm width	Could have		
The chair provides sustained seating comfort	Maximum continous sitting time without discomfort	≥ 30 minutes (user-reported comfort ≥ 4/5)	Should have		
The chair provides a physically comfortable and emotionally reassuring experience	User comfort rating	≥ 4/5 in testing	Could have		
The chair contributes to a calm, safe, and positive bathroom atmosphere	Noise level during operation	≤ 30 dB	Could have		
The chair enabled intuitive understanding of its technical aspects	Number of steps to perform core tasks	≤ 3 steps for adjust, ≤ 2 steps for manoeuvre	Should have		
The chair allows operation without prior technical knowledge	Time to learn basic operation	≤ 2 minutes for first-time user	Should have		
The chair supports independent use by one caregiver or user	Required operator count	1 person	Must have		
The chair can reduce cognitive load by simplifying operation	Number of controls	≤ 4 controls, all labeled	Could have		
The chair incorporates a color palette	Number of colorways	≥ 4 color options	Could have		
The chair fits naturally into a home environment	Home integration rating	≥ 4/5 in user testing	Could have		
The chair provides medical functionality without appearing clinical or institutional	Clinical appearance rating	≤ 2/5 in user testing	Could have		
The chair maintains a minimal silhouette while allowing subtle personal expression	Maximum visual volume	≤ 0.12 m^3	Could have		
The chair incorporates optional accent details for users who prefer expressive aesthetics	Number of optional accent details	≥ 3 options	Could have		
The chair incorporates a specific design style	Style adherence rating	≥ 4/5 expert panel	Could have		
The chair visually harmonizes with a specific material and texture	Number of coordinated material/texture palettes	≥ 2 palettes	Could have		

8.2 Appendix 2 – Anthropometric measurements from PeopleSize

PeoplesSize Measurements	German (65+)	Belgian (65-80)	German (40-64)
Feet length	245.5mm		
Feet width		95.3mm	
Buttock to front of Knee, sitting	591mm		
Buttock to back of Knee (Popliteal tendon), sitting	483.5mm		
Back of Knee height (Popliteal tendon), sitting	421.8mm		
Shoulder height (Acromion), sitting	570.5mm		
Shoulder (Acromion) to underside of Elbow, flexed	349mm		
Height of maximum Lumbar curvature, sitting	229.8mm		
Back of Elbow to grip	326.8mm		
Hip breadth	395.8mm		
Neck width, just below level of Adam's apple			130mm
Broadest part of Cheekbones to back of Head, hori'l	123.5mm		
	Mean of male and female (5-95th percentile)		